SingleRAN

IP Active Performance Measurement Feature Parameter Description

Issue Draft A

Date 2025-12-31

HUAWEI TECHNOLOGIES CO., LTD.

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https://securitybulletin.huawei.com/enterprise/en/security-advisory

Contents

1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 03 (2025-12-14).

- Technical changes

Change Description	Parameter Change	Base Station Model
Added support for IP Active Performance Measurement by the UMPThb. For details, see: • **4.2.2.1 TWAMP Controller Function** • **4.2.2.2 TWAMP Responder Function** • **5.3.3 Hardware**	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Added support for TWAMP defined in RFC 5938. For details, see: • 3 Overview • 4.1.1.2 TWAMP Implementation Process • 4.1.1.3 Protocol Compliance • 4.5.1.1 Data Preparation • 4.5.1.2 Using MML Commands • 4.5.2 Activation Verification • 4.5.3 Network Monitoring	Added the **TWAMPCLIENT**.***INDIVSESSCTRLSW (LTE eNodeB, 5G gNodeB)*** parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NR, and NB-IoT.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

RAT	Feature ID	Feature Name	Chapter/Section
GSM	GBFD-151201	BSC IP Active Performance Measurement	4 TWAMP
GSM	GBFD-151202	BTS IP Active Performance Measurement	
UMTS	WRFD-151211	RNC IP Active Performance Measurement	
UMTS	WRFD-151212	NodeB IP Active Performance Measurement	
LTE FDD	LOFD-070219	IP Active Performance Measurement	
LTE TDD	TDLOFD-003018	IP Active Performance Measurement	
NB-IoT	MLOFD-070219	IP Active Performance Measurement	
NR	FOFD-010060	Transmission Network Detection and Reliability Improvement	

The GBTS does not support the IP Active Performance Measurement feature.

The Two-Way Active Measurement Protocol (TWAMP) has defined a security protection mechanism, which is not supported in this version. However, security protection can be implemented for TWAMP by implementing IPsec at the IP layer or by adding access control list (ACL) rules.

3 Overview

IP performance monitoring (IP PM)–related standards defined by the Internet Engineering Task Force (IETF) support configuration and maintenance of IP-based transport networks and improve the interoperability between transmission devices and test instruments and the testability of IP performance on end-to-end (E2E) Ethernet links.

Huawei introduced the IP Active Performance Measurement feature in accordance with the IETF IP PM standards listed in the following table.

Protocol Number	Protocol Name
RFC 5357	A Two-way Active Measurement Protocol (TWAMP)
RFC 5938	Individual Session Control Feature for the Two-Way Active Measurement Protocol
RFC 2680	A One-way Packet Loss Metric for IPPM
RFC 2681	A Round-trip Delay Metric for IPPM
RFC 3393	IP Packet Delay Variation Metric
RFC 8545	OWAMP/TWAMP Well-Known Ports
RFC 2544	Benchmarking Methodology for Network Interconnect Devices

TWAMP supports IP performance measurement on connections between network elements (NEs) and devices that support TWAMP in a radio transport network. The performance metrics include one-way or two-way packet loss rate, round-trip delay, and one-way delay variation, which will be elaborated in **4.1.2 Measurement Parameters**. IP performance measurement can be implemented on connections over the X2 interface between eNodeBs, over the X2 interface between an eNodeB and a gNodeB, over the Xn interface between two gNodeBs, between a GSM/UMTS dual-mode base station and a base station controller, between an eNodeB/gNodeB and a core network (CN) device, between base station controllers, between a base station controller and a CN device, between an

NE in a radio network and a transmission device (for example, a router), and between an NE in a radio network and a test device.

In addition, a Huawei base station can serve as a reflector to enable the peer end to measure the transport network quality in accordance with RFC 2544.

RFC 2544 tests are performed between an NE in a radio network (as a reflector) and an RFC 2544–compliant initiator (for example, a router) or a tester in a radio transport network to monitor the performance of transmission links. The performance metrics include throughput, two-way delay, and two-way packet loss rate.

4 TWAMP

4.1 Principles

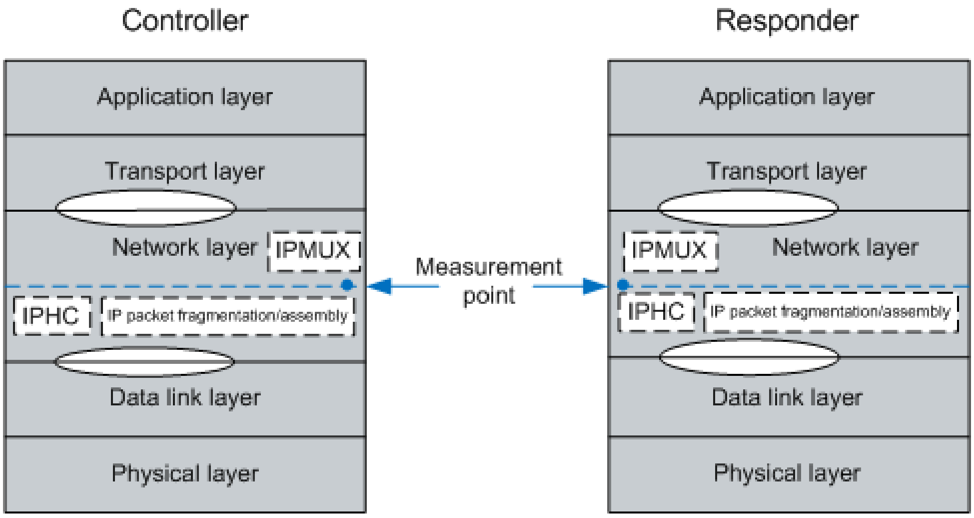
TWAMP supports IP performance measurement on connections between network elements (NEs) and devices that support TWAMP in a radio transport network. The performance metrics include one-way or two-way packet loss rate, round-trip delay, and one-way delay variation, which will be elaborated in **4.1.2 Measurement Parameters**. IP performance measurement can be implemented on connections over the X2 interface between eNodeBs, over the X2 interface between an eNodeB and a gNodeB, over the Xn interface between two gNodeBs, between a GSM/UMTS dual-mode base station and a base station controller, between an eNodeB/gNodeB and a core network (CN) device, between base station controllers, between a base station controller and a CN device, between an NE in a radio network and a transmission device (for example, a router), and between an NE in a radio network and a test device.

In accordance with the TWAMP protocol, TWAMP measures the transmission quality at the IP layer. The controller sends test packets before performing operations such as IP packet fragmentation. The responder performs IP packet assembly before responding to the received test packets. The controller uses TWAMP protocol RFC5357. It is recommended that the TWAMP protocol used on the responder side be the same as that used on the controller side.

Figure 4-1 illustrates the position of TWAMP in the TCP/IP protocol stack.

TWAMP resides above IP packet fragmentation and assembly at the network layer, as shown in **Figure 4-1**.

Figure 4-1 Position of TWAMP in the TCP/IP protocol stack

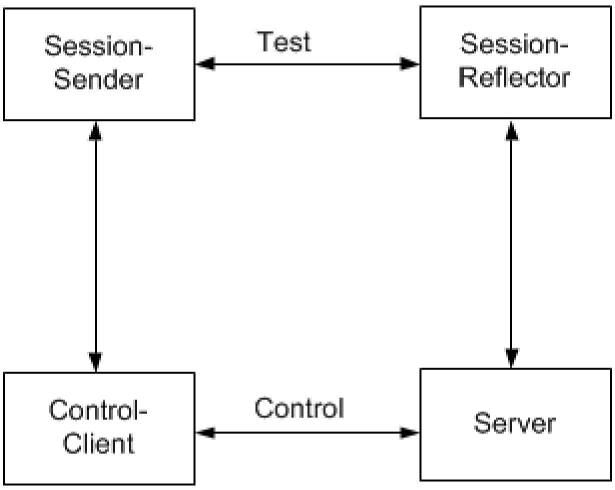


4.1.1 Basic Concepts

4.1.1.1 Logical Entities

TWAMP defines four logical entities: Session-Sender, Session-Reflector, Control-Client, and Server, as shown in **Figure 4-2**. TWAMP also defines two packet types: control packet and test packet.

Figure 4-2 TWAMP logical entities



The following table describes the functions of these logical entities.

Control Entity or Test Entity	Logical Entity	Function
Control entity	Control-Client	The Control-Client and Server exchange control packets to initiate, start, and stop TWAMP test sessions.
	Server	

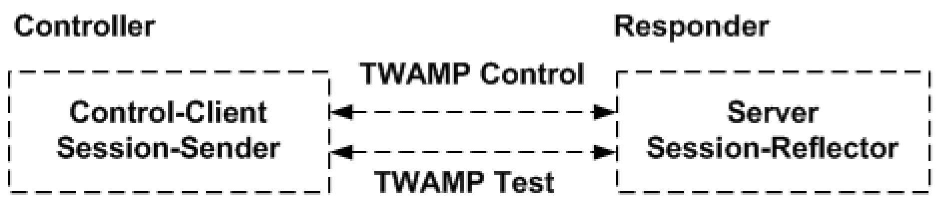
Control Entity or Test Entity	Logical Entity	Function
Test entity	Session-Sender	The Session-Sender sends test packets to the Session-Reflector, which responds to test packets.
	Session-Reflector	

Different TWAMP logical entities can be deployed on the same host in full or light architecture.

Full Architecture

In full architecture, one host (Controller) plays the roles of the Session-Sender and Control-Client, and the other host (Responder) plays the roles of the Session-Reflector and Server, as shown in **Figure 4-3**. The Controller sends TCP-type TWAMP control packets to the Responder for establishing test sessions. After the sessions are established, the Controller sends UDP-type TWAMP test packets to the Responder. The Session-Reflector of the Responder responds to the test packets.

Figure 4-3 TWAMP full architecture



Light Architecture

In light architecture, one host (Controller) plays the roles of the Session-Sender, Control-Client, and Server, and the other host (Responder) plays the role of the Session-Reflector, as shown in **Figure 4-4**. The Controller establishes test sessions with the Server in a non-standard way. Then test packets are transmitted between the Controller and Responder.

The light architecture can be further divided into the stateless mode and stateful mode depending on how the Session-Reflector responds to packets. The light architecture is specified by the **TWAMPSENDER**.***LIGHTARCHITECTUREMODE (5G gNodeB, LTE eNodeB)*** parameter.

The stateful mode is recommended when TWAMP is deployed between Huawei base stations in light architecture. When TWAMP is deployed between Huawei base stations and third-party equipment in light architecture, the stateful mode is preferred as long as the peer equipment supports this mode. If the peer equipment does not support the stateful mode, the stateless mode is used. The following table compares the two architectures.

Item	Full Architecture	Light Architecture
Reliability	This architecture provides a control plane management mechanism to improve reliability.	No control plane management mechanism is available, and the reliability is weaker than that of the full architecture.
Deployment flexibility	This architecture has high requirements on device capabilities, and is therefore less flexible in deployment than the light architecture.	This architecture has low requirements on device capabilities and is flexible in deployment.
Application scenarios	The TWAMP architecture needs to be selected based on the capability of the peer device.	

Figure 4-4 TWAMP light architecture

NOTE

- The TWAMP light architecture is mainly used for IP fault demarcation. It is recommended that the light architecture be used only when the peer end does not support the full architecture and that TWAMP be deactivated after the fault demarcation is complete.
- During TWAMP measurement, one end serves as the TWAMP Controller while the other end serves as the TWAMP Responder. In addition, the TWAMP deployment mode at the two ends must be consistent. If TWAMP light architecture is used, the two ends must be deployed in the same mode.

4.1.1.2 TWAMP Implementation Process

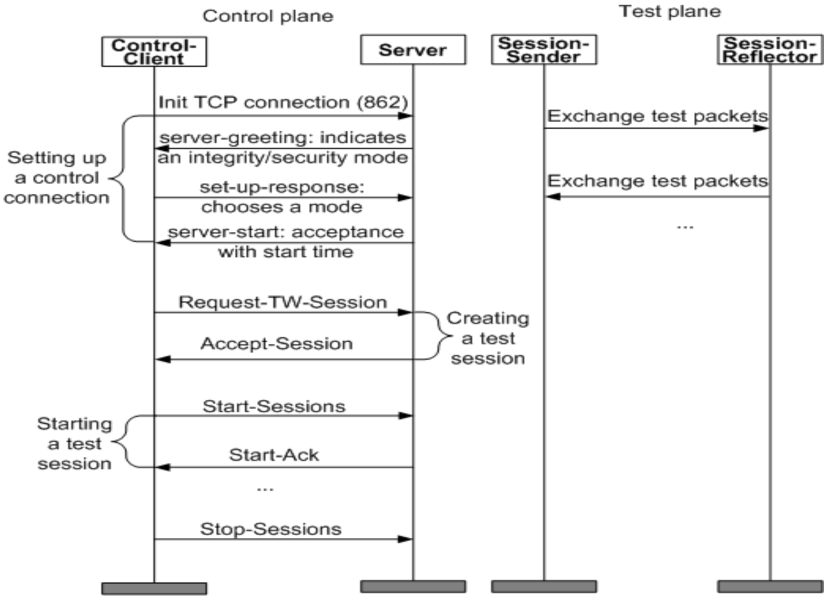
The standard TWAMP implementation process varies depending on the TWAMP architecture. The TWAMP implementation process consists of negotiation and testing in full architecture and includes only testing in light architecture.

Full Architecture

Negotiation is conducted prior to testing between the Control-Client and Server using TCP packets. The Server uses port 862 as the listening port by default, and the listening port is configurable. This port is specified by the **TWAMPCLIENT**.***PEERPORT (5G gNodeB, LTE eNodeB)*** parameter.

Testing is conducted between the Session-Sender and Session-Reflector based on UDP. The ports used by UDP are assigned and managed inside each NE, and they are negotiated between NEs in the negotiation phase, as shown in **Figure 4-5** and **Figure 4-6**.

Figure 4-5 TWAMP measurement (defined in RFC 5357)



TWAMP measurement includes four phases: 1, 2, 3, and 4.

1. Phase 1: Establishing a TCP connection
   a. The Control-Client initiates a TCP connection to the Server on the listening port of the Server.
   b. The Server responds with a server greeting message, indicating the mode of communication it supports.
   c. However, if the Server does not respond or responds with an unexpected mode, the Control-Client can close the connection. Otherwise, the Control-Client sends a set-up-response message to choose a working mode.
   d. The Server responds with a server-start message. The connection setup is complete.
2. Phase 2: Creating test sessions

   The following commands are available for the Control-Client: Request-TW-Session, Start-Sessions, and Stop-Sessions. The Server responds to these commands by sending one of the following messages: Accept-Session or Start-Ack.
   a. The Control-Client sends a Request-TW-Session message to negotiate with the Server about the UDP transmit port number, UDP receive port number, IP address, and differentiated services code point (DSCP). The Control-Client also uses this message to indicate the start time of the test.
   b. The Server responds with an Accept-Session message, indicating that it accepts the negotiated results. The Server can respond with a different UDP port number for the Control-Client to use. The Control-Client receives the port number and enters 3.
3. Phase 3: Starting test sessions
   a. The Control-Client sends a Start-Sessions message, indicating that it starts a test session.

b. The Server responds with a Start-Ack message, indicating that it accepts the test session.

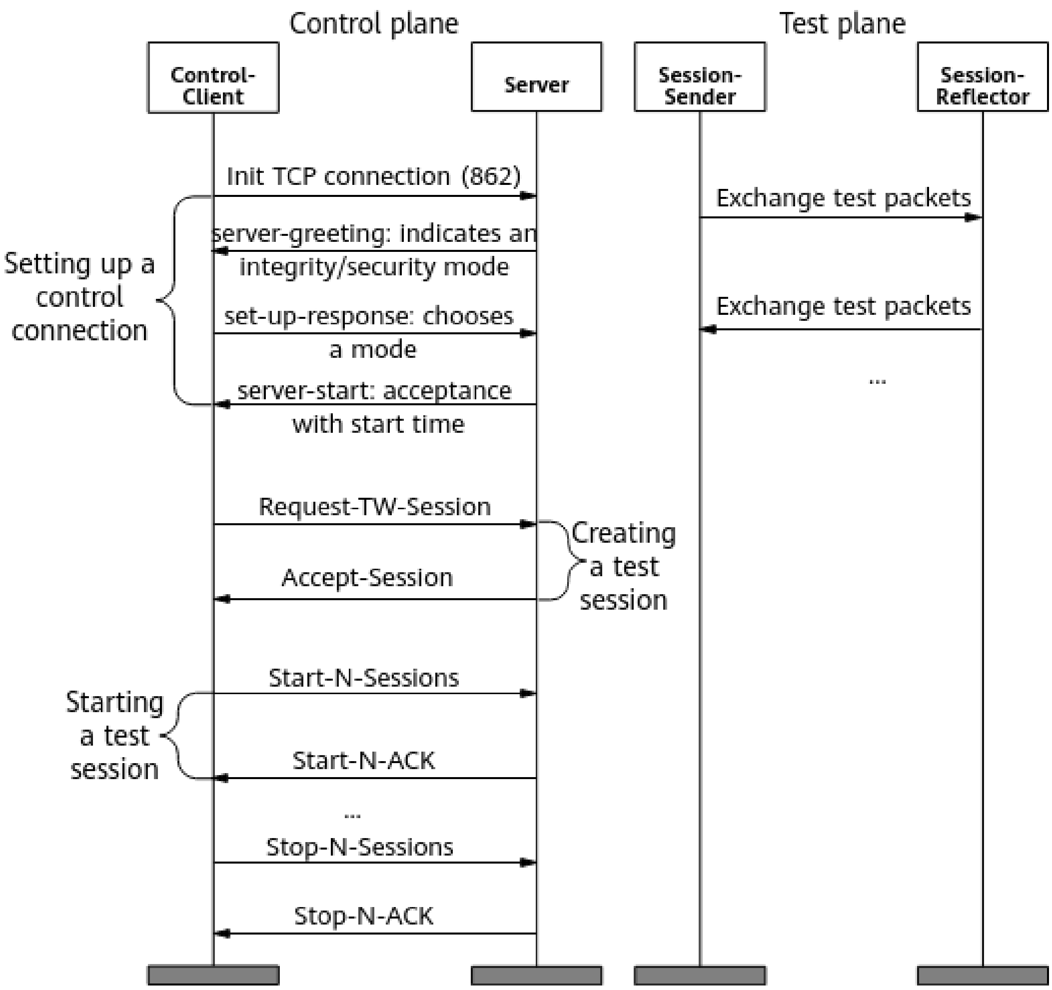
4. Phase 4: Testing

   UDP packets are used for testing. The Session-Sender actively sends test packets to the Session-Reflector in a fixed stream with the same source IP address, destination IP address, source UDP port number, destination UDP port number, and DSCP.

   The test packets are transmitted in unauthenticated mode, authenticated mode, or encrypted mode. In compliance with RFC 5357, only unauthenticated mode can be used.

   – In unauthenticated mode, neither shared keys nor hashed message authentication code (HMAC) keys are used.
   – In authenticated mode, the public key is used.
   – In encrypted mode, negotiation packets and test packets are encrypted.

Figure 4-6 TWAMP measurement (defined in RFC 5938)



TWAMP measurement includes four phases: 1, 2, 3, and 4.

1. Phase 1: Establishing a TCP connection

a. The Control-Client initiates a TCP connection to the Server on the listening port of the Server.

b. The Server responds with a server greeting message, indicating the mode of communication it supports.

c. However, if the Server does not respond or responds with an unexpected mode, the Control-Client can close the connection. Otherwise, the Control-Client sends a set-up-response message to choose a working mode.

d. The Server responds with a server-start message. The connection setup is complete.

2. Phase 2: Creating test sessions

The following commands are available for the Control-Client: Request-TW-Session, Start-N-Sessions, and Stop-N-Sessions. The Server responds to these commands by sending one of the following messages: Accept-Session, Start-N-Ack, and Stop-N-Ack.

a. The Control-Client sends a Request-TW-Session message to negotiate with the Server about the UDP transmit port number, UDP receive port number, IP address, and DSCP. The Control-Client also uses this message to indicate the start time of the test.

b. The Server responds with an Accept-Session message, indicating that it accepts the negotiated results. The Server can respond with a different UDP port number for the Control-Client to use. The Control-Client receives the port number and enters **3**.

3. Phase 3: Starting test sessions

a. The Control-Client sends a Start-N-Sessions message, indicating that it starts a test session.

b. The Server responds with a Start-Ack message, indicating that it accepts the test session.

4. Phase 4: Testing

UDP packets are used for testing. The Session-Sender actively sends test packets to the Session-Reflector in a fixed stream with the same source IP address, destination IP address, source UDP port number, destination UDP port number, and DSCP.

The test packets are transmitted in individual session control mode. In this mode, individual control on one or more test sessions can be performed.

Light Architecture

The TWAMP implementation process includes only testing in light architecture. Testing is conducted based on UDP, and the ports used by UDP is user-defined. The testing phase is the same as phase 4 of TWAMP measurement in full architecture.

4.1.1.3 Protocol Compliance

The compliance with RFC 5357, RFC 5938, and RFC 8545 is described in **Table 4-1** and **Table 4-2**.

Table 4-1 Protocol compliance in IPv4 networking

Product	Full Architecture	Light Architecture
Base stations	Partial compliance with RFC 5357. ● Supports the TWAMP Controller and TWAMP Responder functions. ● Supports only test sessions in unauthenticated mode. ● Supports timestamps in the Coordinated Universal Time (UTC) format in test packets, and does not support real-time synchronization and one-way delay measurement. ● Supports only the filling of 0s in the redundant payload of test sessions, and does not support the filling of pseudo random code. Partial compliance with RFC 5938. ● Supports the TWAMP Controller and TWAMP Responder functions. ● Supports test sessions in individual session control mode. ● Supports individual control on one or more test sessions using SIDs in individual session control mode. Partial compliance with RFC 8545. ● The Control-Client uses the TWAMP-Test well-known port number 862 in the Receiver Port field of the Request-TW-Session message. ● The Session-Reflector responds to test sessions on port 862.	Partial compliance with RFC 5357. ● Supports the TWAMP Controller and TWAMP Responder functions, and supports the stateless and stateful modes. ● Supports only test sessions in unauthenticated mode. ● Supports timestamps in the UTC format in test packets, and does not support real-time synchronization and one-way delay measurement. ● Supports only the filling of 0s in the redundant payload of test sessions, and does not support the filling of pseudo random code.

Product	Full Architecture	Light Architecture
Base station controllers	Partial compliance with RFC 5357. ● Supports the TWAMP Controller and TWAMP Responder functions. ● Supports only test sessions in unauthenticated mode. ● Uses the relative time as the timestamp and encapsulates the timestamp in test packets. ● Supports only the filling of 0s in the redundant payload of test sessions, and does not support the filling of pseudo random code.	Not supported

Table 4-2 Protocol compliance in IPv6 networking

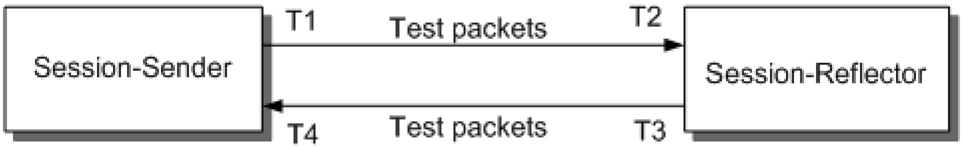
Product	Full Architecture	Light Architecture
Base stations	Partial compliance with RFC 5357. ● Supports the TWAMP Controller and TWAMP Responder functions. ● Supports only test sessions in unauthenticated mode. ● Uses the relative time as the timestamp and encapsulates the timestamp in test packets. ● Supports only the filling of 0s in the redundant payload of test sessions, and does not support the filling of pseudo random code. Partial compliance with RFC 5938. ● Supports the TWAMP Controller and TWAMP Responder functions. ● Supports test sessions in unauthenticated mode and individual session control mode. ● Supports individual control on one or more test sessions using SIDs in individual session control mode. Partial compliance with RFC 8545. ● The Control-Client uses the TWAMP-Test well-known port number 862 in the Receiver Port field of the Request-TW-Session message. ● The Session-Reflector responds to test sessions on port 862.	Partial compliance with RFC 5357. ● Supports the TWAMP Controller and TWAMP Responder functions, and supports the stateless and stateful modes. ● Supports only test sessions in unauthenticated mode. ● Supports timestamps in the UTC format in test packets, and does not support real-time synchronization and one-way delay measurement. ● Supports only the filling of 0s in the redundant payload of test sessions, and does not support the filling of pseudo random code.

4.1.2 Measurement Parameters

TWAMP actively inserts test packets into links being tested and calculates the packet loss rate, round-trip delay, and delay variation based on the fields contained in the test packets.

Figure 4-7 shows the test process.

Figure 4-7 Test process



The detailed process is as follows:

1. The Session-Sender sends test packets containing sequence numbers and timestamp T1.
2. The Session-Reflector records timestamp T2 upon receiving the test packets from the Session-Sender. The Session-Reflector then copies the sequence numbers and timestamp T1 into the response packets and also includes the Session-Reflector's transmit sequence numbers and timestamp T3 in the response packets.

   In the case of TWAMP light architecture, the Session-Reflector can work in either the stateful or stateless mode.

   – In stateful mode, the sequence numbers filled in by the Session-Reflector are generated incrementally based on session. This mode enables the measurement of the forward packet loss rate and backward packet loss rate on the peer device.
   – In stateless mode, the sequence numbers filled in by the Session-Reflector are copied from the sequence numbers of received packets. This mode enables the measurement of two-way packet loss rate on the peer device.
3. Upon receiving the response packets, the Session-Sender records timestamp T4 and calculates the number of received packets.

The Session-Sender periodically sends packets at an interval of 10 ms to 1000 ms. The interval can be a fixed value or a random value, depending on the setting of the **TWAMPSENDER**.***PKTINTTYPE (LTE eNodeB, 5G gNodeB)*** parameter. When the packets need to be sent at a fixed interval, you also need to set the **TWAMPSENDER**.***PKTINT (LTE eNodeB, 5G gNodeB)*** parameter to the desired value.

NOTE

- According to the TWAMP, the Session-Reflector returns a packet at the earliest opportunity.
- TWAMP defines negotiation timeout and test timeout for the Responder, which are specified by the ***SERVWAIT (LTE eNodeB, 5G gNodeB)*** and ***REFWAIT (LTE eNodeB, 5G gNodeB)*** parameters, respectively.
  - ***SERVWAIT (LTE eNodeB, 5G gNodeB)***: The Server closes the connection during the negotiation if it has not received any negotiation packet within the period specified by this parameter. This parameter is configurable with the default value of **900** (unit: s). This parameter takes effect only when the TWAMP Responder is deployed in full architecture.
  - ***REFWAIT (LTE eNodeB, 5G gNodeB)***: During a test, the Session-Reflector releases resources if it has not received any test packets within the period specified by this parameter, and the Server closes the connection. This parameter is configurable with the default value of **900** (unit: s). When serving as the Controller, a base station or base station controller reinitiates a negotiation if it has not received any test packets within 900s during the test. This occurs because the Controller assumes that the peer end may have deleted the session.
  - When the TWAMP Responder works in stateful mode, it does not respond to the first packet of session self-learning.

4.1.2.1 Packet Loss Rate

The packet loss rate indicates the transmission quality of a tested IP link. This feature performs packet loss measurement. Based on the number of packets transmitted and received by the Session-Sender and the number of packets transmitted and received by the Session-Reflector, the packet loss rate is calculated by dividing the number of lost packets by the total number of transmitted packets.

The calculation formulas are as follows:

Forward packet loss rate = (Number of packets transmitted by the Session-Sender – Number of packets received by the Session-Reflector)/Number of packets transmitted by the Session-Sender

Backward packet loss rate = (Number of packets transmitted by the Session-Reflector – Number of packets received by the Session-Sender)/Number of packets transmitted by the Session-Reflector

Two-way packet loss rate = (Number of packets transmitted by the Session-Sender – Number of packets received by the Session-Sender)/Number of packets transmitted by the Session-Sender

NOTE

- Forward indicates the direction from Session-Sender to Session-Reflector. The forward packet loss rate is valid only when the Session-Sender works in full architecture or in stateful mode of the light architecture.
- Backward indicates the direction from Session-Reflector to Session-Sender. The backward packet loss rate is valid only when the Session-Sender works in full architecture or in stateful mode of the light architecture.
- If the Session-Sender does not receive response packets due to severe forward or backward packet losses on the transport network, the NE considers that all packets in the forward direction are lost.
- The two-way packet loss rate is valid only when the Session-Sender works in stateless mode of the light architecture.
- In normal cases, the number of packets received by the Session-Reflector equals the number of packets transmitted by the Session-Reflector.

4.1.2.2 RTT

The RTT is the length of time it takes for a packet to be sent plus the length of time it takes for an acknowledgment of that packet to be received. The RTT indicates the delay in a transmission network.

This feature calculates the RTT based on the sending time of the packet at the Sender (T1), receiving time of the packet at the Reflector (T2), sending time of the response packet at the Reflector (T3), and receiving time of the response packet at the Sender (T4).

The formula is as follows:

RTT = (T2 – T1) + (T4 – T3) = (T4 – T1) – (T3 – T2)

4.1.2.3 Delay Variation

The delay variation indicates the difference between delays of selected packets on an IP link. A greater difference in the delay results in a larger delay variation.

This feature calculates the delay variation based on the delays of two adjacent test packets.

Forward delay variation indicates the difference between the forward delays of two adjacent test packets.

Backward delay variation indicates the difference between the backward delays of two adjacent test packets.

NOTE

TWAMP results may be inaccurate during router switchovers and active/standby Ethernet port switchovers.

TWAMP supports active/standby board switchovers. If the Controller experiences an active/standby board switchover, it reinitiates a negotiation and restarts a test until the negotiation is successful. If the Responder experiences an active/standby board switchover, it will not respond to any tests, and the ongoing TWAMP test will be affected.

4.2 TWAMP Application

4.2.1 Differences Between TWAMP and Huawei-Proprietary IP PM

TWAMP applies to the following interfaces:

- GSM interfaces: Abis, A, and Gb
- UMTS interfaces: Iub, Iu, Iur, and uX2
- LTE interfaces: S1, X2, and eX2
- NR interfaces: S1, X2, NG, eXn, and Xn

The working principles of TWAMP on these interfaces are basically the same. For details, see **4.1 Principles**.

The base station and base station controller's TWAMP-related activation commands and configuration parameters as well as measurement parameters that support TWAMP are described in **4.2.2 TWAMP Application on the Base Station Side** and **4.2.3 TWAMP Application on the Base Station Controller Side**.

NOTE

TWAMP application on the co-MPT multimode base station is not discussed in this document because it is the same as TWAMP application on the eGBTS, NodeB, eNodeB, and gNodeB.

The following paragraphs explain the differences between TWAMP and Huawei-proprietary IP PM from technical and application perspectives. For details about Huawei-proprietary IP PM, see *IP Performance Monitor*.

Technical Differences

TWAMP is defined by the IETF while Huawei IP PM is a proprietary protocol of Huawei.

Table 4-3 Technical differences between TWAMP and Huawei-proprietary IP PM

Item	Huawei-Proprietary IP PM	TWAMP
Interconnection	The peer device must be provided by Huawei.	The peer device can be provided by any vendors as long as it supports TWAMP.
Test	Service packets are measured.	Injected test packets are measured. Offline measurement is recommended.

<table>
<tr><th>Item</th><th>Huawei-Proprietary IP PM</th><th>TWAMP</th></tr>
<tr><td>Restriction</td><td>Huawei-proprietary IP PM applies only to online services.</td><td>● Packet injection affects ongoing services and occupies bandwidth.
● The peer device must support TWAMP.</td></tr>
</table>

NOTE

Huawei-proprietary IP PM uses the combination of source IP address, destination IP address, UDP, and DSCP or the combination of source IP address, destination IP address, and UDP to identify a transmission link. TWAMP uses the combination of source IP address, destination IP address, source UDP port number, destination UDP port number, and DSCP to identify a transmission link.

Application Differences

Both TWAMP and Huawei-proprietary IP PM measure the transmission quality in real time. Huawei-proprietary IP PM is recommended if both ends use Huawei devices, for example, between a Huawei base station and a Huawei base station controller, between a Huawei eNodeB and a Huawei S-GW, and between two Huawei base stations. TWAMP is recommended if devices used at the two ends are provided by different vendors, for example, between a Huawei base station or base station controller and a transmission device provided by another vendor.

TWAMP and Huawei-proprietary IP PM are complementary. Only TWAMP is available in the following scenarios:

- Transmission counters if there are no running services, for example, services cannot be accessed in the case that the base station is disconnected but in good transmission condition, or if there are no running services or the service load is light in the early morning
- Both local and peer devices support TWAMP

Table 4-4 Application differences between TWAMP and Huawei-proprietary IP PM

<table>
<tr><th>Item</th><th>Scenario</th><th>Huawei-Proprietary IP PM</th><th>TWAMP</th></tr>
<tr><td rowspan="2">Maintenance and testing</td><td>When there are no ongoing services</td><td>Not supported</td><td>Supported</td></tr>
<tr><td>Lub/Abis connections between the base station and base station controller</td><td colspan="2">Huawei-proprietary IP PM is recommended.</td></tr>
</table>

Item	Scenario	Huawei-Proprietary IP PM	TWAMP
	Connections between a RAN device and an intermediate transmission device	Not supported	Supported
	QoS measurement of S1/X2/eX2 connections between an eNodeB and a Huawei S-GW/eNodeB in LTE	Huawei-proprietary IP PM is recommended.	
	QoS measurement of S1/X2 connections between an eNodeB and a non-Huawei S-GW/eNodeB	Not supported	Supported
	QoS measurement of X2 connections between a Huawei gNodeB and a Huawei eNodeB in non-standalone NR (NSA NR)	Huawei-proprietary IP PM is recommended.	
	QoS measurement of Xn/eXn connections between two gNodeBs in SA NR	Huawei-proprietary IP PM is recommended.	
	QoS measurement of S1 and NG connections between a Huawei gNodeB and a Huawei CN	Not supported	Supported
Monitoring	Online monitoring of wireless services on the Iub/uX2/Abis/S1/X2/eX2/Xn/eXn interface	Supported	Not supported
	QoS monitoring of transmission between a base station/base station controller and a router or a non-Huawei S-GW	Not supported	Supported

4.2.2 TWAMP Application on the Base Station Side

This section describes the TWAMP application on the base station side. The related features are as follows:

- GSM: GBFD-151202 BTS IP Active Performance Measurement
- UMTS: WRFD-151212 NodeB IP Active Performance Measurement
- LTE: LOFD-070219 IP Active Performance Measurement
- LTE: TDLOFD-003018 IP Active Performance Measurement

- LTE: MLOFD-070219 IP Active Performance Measurement
- NR: FOFD-010060 Transmission Network Detection and Reliability Improvement

When TWAMP is applied on a base station, the roles of the base station vary depending on the connected peer device.

Table 4-5 Base station's roles with TWAMP

Peer Device	Base Station's Role
Transmission device	The base station is configured as the TWAMP Controller because most transmission devices can work as the TWAMP Responder.
CN	When a CN is used, it is recommended that the CN and eNodeB be configured as the TWAMP Controller and TWAMP Responder, respectively.
Base station controller	Huawei-proprietary IP PM is preferred when there are ongoing services. TWAMP is used when there is no service. It is recommended that the base station controller and base station be configured as the TWAMP Controller and TWAMP Responder, respectively.
Test device	When a test device, such as Spirent TestCenter, supports TWAMP, the test device and base station are configured as the TWAMP Controller and TWAMP Responder, respectively.
Peer base station	Generally, the local base station is configured as the TWAMP Controller, and the peer base station is configured as the TWAMP Responder.

4.2.2.1 TWAMP Controller Function

When the base station serves as the TWAMP Controller, it is responsible for initiating tests, collecting statistics, and displaying test results at the local end. The peer end, for example, a router, base station controller, or base station, serves as the Responder, which responds to the negotiation and test packets.

A base station configured with the UMPTb/UMDU supports 16 Control-Clients, each supporting a maximum of 16 Session-Sender test streams; the base station supports 16 Session-Sender test streams in total. A base station configured with the UMPTe/UMPTg/UMPTga/UMPTha/UMPThb supports 24 Control-Clients, each supporting a maximum of 16 Session-Sender test streams; the base station supports 32 Session-Sender test streams in total. A test can be activated by running the **ADD TWAMPCLIENT** and **ADD TWAMPSENDER** commands.

4.2.2.2 TWAMP Responder Function

When the base station serves as the TWAMP Responder, it passively responds to test packets. The base station controller or base station at the peer end serves as the TWAMP Controller, which initiates tests.

A base station configured with the UMPTb/UMDU supports 16 Responders, each supporting a maximum of 16 passive response test streams; each board supports 16 passive response test streams in total. A base station configured with the UMPTe/UMPTg/UMPTga/UMPTha/UMPThb supports 16 Responders, each supporting a maximum of 32 passive response test streams; each board supports 32 passive response test streams in total. The TWAMP Responder function can be activated by running the **ADD TWAMPRESPONDER** command.

4.2.2.3 Networking Scenarios

Table 4-6 describes the networking scenarios where a base station supports TWAMP.

Table 4-6 Networking scenarios where a base station supports TWAMP

Networking Scenario		Supported or Not	Description
Port level	FE/GE/10GE/25GE ports as the transmission ports	Supported	N/A
	E1/T1 ports as the transmission ports	Supported	E1/T1 ports are provided by the UMPT or UMDU boards.
	UMPT interconnection ports	Supported	N/A
Link level	Active and standby routes	Supported	N/A
Board level	UTRPc as the transmission interface board in a multimode base station	Supported	It is recommended that TWAMP be enabled on the UTRPc board.
Site level	Cascaded base stations	Supported	TWAMP can be used on all cascaded base stations.

4.2.3 TWAMP Application on the Base Station Controller Side

This section describes the TWAMP application on the base station controller side. The related features are as follows:

- GSM: GBFD-151201 BSC IP Active Performance Measurement
- UMTS: WRFD-151211 RNC IP Active Performance Measurement

When TWAMP is applied on a base station controller, the roles of the base station controller differ depending on the connected peer device, as described in **Table 4-7**.

Table 4-7 Base station controller's roles with TWAMP

Peer Device	Base Station Controller's Role
Transmission device	The base station controller is configured as the TWAMP Controller because most transmission devices support the TWAMP Responder function.
Base station	Huawei-proprietary IP PM is preferred when the base station is provided by Huawei. TWAMP is used when there is no service or when the base station is provided by another vendor. It is recommended that the base station and base station controller be configured as the TWAMP Responder and TWAMP Controller, respectively.

4.2.3.1 TWAMP Controller Function

When the BSC/RNC serves as the TWAMP Controller, it is responsible for actively initiating tests, collecting statistics, and displaying test results at the local end. The peer end, for example, a router, a base station controller, a base station, or CN equipment serves as the Responder, which responds to the negotiation and test packets.

The base station controller BSC6900/BSC6910 supports 1024 Control-Clients, each supporting a maximum of 16 Session-Sender test streams. And the BSC6900/BSC6910 supports 1024 Session-Sender test streams in total.

The base station controller BSC6960 supports 24 Control-Clients, each supporting a maximum of 16 Session-Sender test streams. And the BSC6960 supports 32 Session-Sender test streams in total.

A test can be activated by running the **ADD TWAMPCLIENT** and **ADD TWAMPSENDER** commands.

4.2.3.2 TWAMP Responder Function

When the BSC/RNC serves as the TWAMP Responder, it passively responds to test packets. The peer end, for example, a base station controller, a base station, a test device, or CN equipment serves as the TWAMP Controller, which actively initiates tests. The local end passively responds to test packets.

The base station controller BSC6900/BSC6910 supports 1024 Responders, each supporting a maximum of 160 passive response test streams. Each board of the BSC6900/BSC6910 supports a maximum of 160 passive response test streams.

The base station controller BSC6960 supports 16 Responders, each supporting a maximum of 32 passive response test streams. The BSC6960 supports a maximum of 32 passive response test streams.

A passive response test can be activated by running the **ADD TWAMPRESPONDER** command.

4.2.3.3 Networking Scenarios

Table 4-8 describes whether TWAMP is supported on the base station controller side in different networking scenarios.

Table 4-8 Whether TWAMP is supported on the base station controller side in different networking scenarios

<table>
<tr><th colspan="2">Networking Scenario</th><th>Supported or Not</th><th>Description</th></tr>
<tr><td rowspan="2">Port level</td><td>FE/GE ports as the transmission ports</td><td>Supported</td><td>The FG2c/FG2d/FG2e/GOUc/GOUe/GOUd/GOUf/EXOUa/EXOUb/CMPU board supports TWAMP.</td></tr>
<tr><td>IP over E1/T1 applied on the RNC</td><td>Not supported</td><td>N/A</td></tr>
<tr><td rowspan="4">Link level</td><td>Active and standby link aggregation groups (LAGs)</td><td>Supported</td><td>N/A</td></tr>
<tr><td>LAGs working in load sharing mode</td><td>Supported</td><td>N/A</td></tr>
<tr><td>Routes working in load sharing mode</td><td>Supported</td><td>N/A</td></tr>
<tr><td>Active and standby routes</td><td>Supported</td><td>N/A</td></tr>
<tr><td rowspan="2">Board level</td><td>Active and standby boards</td><td>Supported</td><td>N/A</td></tr>
<tr><td>Interface boards working in transmission resource pool mode</td><td>Supported</td><td>An IP address is assigned at the local end for starting TWAMP measurement.</td></tr>
</table>

4.3 Network Analysis

4.3.1 Benefits

TWAMP reduces O&M cost. Specifically, it provides the following benefits:

- Quick transport network performance monitoring

  If the transmission rate is unstable and the transmission bandwidth changes, this feature enables operators to monitor the QoS of the transport network, and to quickly identify transmission network problems for future capacity expansion and network optimization.
- Quick fault diagnosis and low maintenance costs
  - This feature uses TWAMP to quickly troubleshoot transmission faults, such as high packet loss rate or long delay.
  - This feature enables troubleshooting a transport network by segment, which improves network maintainability and reduces maintenance costs.

However, TWAMP uses User Datagram Protocol (UDP) packet injection, which generates traffic and occupies bandwidth. For example, if 80-byte packets are continuously sent at a rate of 10 packets per second in a test stream, a bandwidth of 6.4 kbit/s is consumed.

4.3.2 Impacts

The impacts between this function and other functions are described in **4.4.2.3 Function Impacts**.

- Because TWAMP negotiation packet interaction occurs in the protocol stack and only a small number of packets are involved, TWAMP has negligible impact on CPU performance.
- TWAMP test packets affect the user-plane forwarding performance because they are transmitted and received on the user plane. The greater the transmit rate of test packets, the greater the resource consumption of TWAMP forwarding. However, TWAMP forwarding resource consumption still has negligible impact when compared with the base station and base station controller's forwarding capabilities.
- TWAMP testing uses packet injection, which generates traffic in the transport network and therefore occupies some bandwidth. The bandwidth consumption is related to the transmit rate of test packets. Users can specify the transmit interval and length for the packets to be transmitted.
- In maintenance and testing scenarios, if you are not sure whether the transmit rate (determined by the IP path, resource group, port, and other factors) on transmission links is close to the planned bandwidth, transmitting a small number of packets at an appropriate interval (low-traffic) is recommended for a TWAMP test. For example, if 80-byte packets are continuously sent at a rate of 10 packets per second in a test stream, the bandwidth consumption is 6.4 kbit/s.
- In monitoring scenarios, it is recommended that you reserve bandwidth for TWAMP tests so that test packets can be sent continuously. If 80-byte packets are sent at a rate of 10 packets per second in a test stream, the bandwidth consumption is 6.4 kbit/s. You can monitor the test stream to check whether any transmission faults are occurring.

4.4 Requirements

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.4.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
GSM	GBFD-151201	BSC IP Active Performance Measurement	LGMIBSCTWAMP	per TRX
GSM	GBFD-151202	BTS IP Active Performance Measurement	LGB3TWAMP01	per BTS
UMTS	WRFD-151211	RNC IP Active Performance Measurement	LQW1RNCTWAMPRESE & LQW1RNCTWAMPRESM	per Erl & Mbps
UMTS	WRFD-151212	NodeB IP Active Performance Measurement	LQW9IPAPM01	per NodeB
LTE FDD	LOFD-070219	IP Active Performance Measurement	LT1S0IPAPM00	per eNodeB
LTE TDD	TDLOFD-003018	IP Active Performance Measurement	LT1SIPAPM000	per eNodeB
NB-IoT	MLOFD-070219	IP Active Performance Measurement	ML1S0IPAPM00	per eNodeB
NR	FOFD-010060	Transmission Network Detection and Reliability Improvement	NR0S0TNDER00	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

4.4.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.4.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
GSM	Abis over IP	None	*IPv4 Transmission*	Applied to the Abis interface of the BSC6900/ BSC6910/ BSC6960
GSM	A over IP	None	*IPv4 Transmission*	Applied to the A interface of the BSC6900
GSM	Gb over IP	None	*IPv4 Transmission*	Applied to the Gb interface of the BSC6900/ BSC6910/ BSC6960
GSM	A over IP Based on Dynamic Load Balancing	None	*Transmission Resource Pool in BSC* in *GBSS Feature Documentation*	Applied to the A interface of the BSC6910/ BSC6960
UMTS	IP Transmission Introduction on Iub Interface	None	*IPv4 Transmission*	Applied to the Iub interface in non-IP pool networking
UMTS	IP Transmission Introduction on Iu Interface	None	*IPv4 Transmission*	Applied to the Iu interface in non-IP pool networking
UMTS	IP Transmission Introduction on Iur Interface	None	*IPv4 Transmission*	Applied to the Iur interface in non-IP pool networking

RAT	Function Name	Function Switch	Reference	Description
UMTS	Iu/Iur Transmission Resource Pool of RNC	None	*Transmission Resource Pool in RNC* in *RAN Feature Documentation*	Applied to the Iu or Iur interface of the BSC6900 in IP pool networking
UMTS	Iub Transmission Resource Pool of RNC	None	*Transmission Resource Pool in RNC* in *RAN Feature Documentation*	Applied to the Iub interface of the BSC6900 in IP pool networking
UMTS	Iub IP Transmission Based on Dynamic Load Balancing	None	*Transmission Resource Pool in RNC* in *RAN Feature Documentation*	Applied to the Iub interface of the BSC6910/ BSC6960 in IP pool networking
UMTS	Iu/Iur IP Transmission Based on Dynamic Load Balancing	None	*Transmission Resource Pool in RNC* in *RAN Feature Documentation*	Applied to the Iu/Iur interface of the BSC6910/ BSC6960 in IP pool networking

4.4.2.2 Mutually Exclusive Functions

None

4.4.2.3 Function Impacts

<table>
<tr><th>RAT</th><th>Function Name</th><th>Function Switch</th><th>Reference</th><th>Description</th></tr>
<tr><td>GSM</td><td>UDP loopback function</td><td>None</td><td>3900 & 5900 Series Base Station MO and Parameter Reference</td><td rowspan="2">If one TWAMP test end is enabled with the UDP loopback function (configured using the SET UDPLOOP command) and a specified IP address or all IP addresses are used for TWAMP testing, test packets will be directly looped back, affecting statistics accuracy.
If the base station functions as the TWAMP Controller, the Responder's response packets will be looped back. As a result, UDP packets are repeatedly transmitted between the local end and peer end (Responder), which consumes more network resources.</td></tr>
<tr><td>UMTS</td><td>UDP loopback function</td><td>None</td><td>3900 & 5900 Series Base Station MO and Parameter Reference</td></tr>
<tr><td>LTE FDD
LTE TDD</td><td>UDP loopback function</td><td>None</td><td>3900 & 5900 Series Base Station MO and Parameter Reference</td><td rowspan="2">If one TWAMP test end is enabled with the UDP loopback function (configured using the SET UDPLOOP command) and a specified IP address or all IP addresses are used for TWAMP testing, test packets will be directly looped back, affecting statistics accuracy.
If the base station functions as the TWAMP Controller, the Responder's response packets will be looped back. As a result, UDP packets are repeatedly transmitted between the local end and peer end (Responder), which consumes more network resources.</td></tr>
<tr><td>NR FDD
Low-frequency NR TDD</td><td>UDP loopback function</td><td>None</td><td>3900 & 5900 Series Base Station MO and Parameter Reference</td></tr>
</table>

4.4.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite
- BSC6900/BSC6910/BSC6960

Boards

NE	Board	Supports This Feature or Not
BSC6900	FG2c/FG2d/FG2e/GOUc/GOUe/GOUd/EXOUa/EXOUb	Yes
BSC6910	FG2c/FG2d/FG2e/GOUc/GOUe/GOUf/GOUd/EXOUa/EXOUb	Yes
BSC6960	CMPU	Yes
Base stations using IPv4 transmission (in TWAMP full architecture)	UMPT/UMDU/UTRPc	Yes
Base stations using IPv4 transmission (in TWAMP light architecture)	UMPT/UMDU/UTRPc	Yes
Base stations using IPv6 transmission (in TWAMP full architecture)	UMPT/UMDU	Yes
Base stations using IPv6 transmission (in TWAMP light architecture)	UMPT/UMDU	Yes

RF Modules

None

4.4.4 Others

- The part of feature implementation undefined in RFC 5357 and its updates complies with the common practice in the industry, and may differ from the feature implementation of some vendors. To ensure the smooth application of this feature, test this feature based on actual network configurations, and enable this feature only after this feature is proved to be normal. If you have any questions about feature implementation, contact Huawei technical support.
- Virtual local area network (VLAN) planning and configuration

  Base station controller: VLAN tags can be added to negotiation and test packets based on the next hop.

  Base station: In single VLAN mode, VLAN tags can be added to negotiation and test packets based on the next hop. In VLAN group mode, negotiation packets use the VLAN of the OM_H configured in the next hop, and test packets use the VLAN of the data packets configured with the same DSCP in the next hop. When a new transmission configuration model is used, that is, when the **GTRANSPARA**.***TRANSCFGMODE (LTE eNodeB, 5G gNodeB)*** parameter is set to **NEW** in VLAN mode, VLAN tags are added to negotiation and test packets based on the interface.

4.5 Operation and Maintenance

4.5.1 Data Configuration

4.5.1.1 Data Preparation

It is recommended that you enable this feature to monitor transmission networks if the bandwidth is sufficient. It is recommended that you only enable this feature temporarily for troubleshooting transmission network faults if the bandwidth of the transmission network is limited when a data service performance fault occurs, such as an unstable download rate. After the troubleshooting is complete, disable this feature.

TWAMP on different NEs is independent of each other. The MML commands, parameters, and data preparation for NEs vary, depending on the deployment architecture and IP version, as described in **Table 4-9**, **Table 4-10**, **Table 4-11**, **Table 4-12**, **Table 4-13**, and **Table 4-14**.

Table 4-9 Data preparation for the base station controller BSC6900/BSC6910 serving as the TWAMP Controller

Parameter Name	Parameter ID	Setting Notes
Local IP Address	**TWAMPCLIENT**.***LOCALIP***	N/A

Parameter Name	Parameter ID	Setting Notes
Peer IP Address	**TWAMPCLIENT**.***PEERIP***	N/A
Peer TCP Port No.	**TWAMPCLIENT**.***PEERPORT***	N/A
Client Index	**TWAMPCLIENT**.***CLIENTID***	N/A
DSCP	**TWAMPCLIENT**.***DSCP***	The default value is **46**.
DSCP	**TWAMPSENDER**.***DSCP***	It is recommended that you set this parameter to the priority of the service packets for which the user shows concern.
Packet Size Type	**TWAMPSENDER**.***PKTSIZETYPE***	The default value **FIXED** is recommended.
Packet Size	**TWAMPSENDER**.***PKTSIZE***	Retain the default value **128**.
Packet Send Interval Type	**TWAMPSENDER**.***PKTINTTYPE***	Retain the default value **FIXED**.
Packet Send Interval	**TWAMPSENDER**.***PKTINT***	N/A

Table 4-10 Data preparation for the base station controller BSC6960 serving as the TWAMP Controller

Parameter Name	Parameter ID	Setting Notes
TWAMP Architecture	**TWAMPCLIENT**.***TWAMPARCH***	Set this parameter to a value consistent with that of the TWAMP responder.
Light Architecture Mode	**TWAMPSENDER**.***LIGHTARCHITECTUREMODE***	Set this parameter to a value consistent with that of the TWAMP responder. Otherwise, the counters related to the packet loss rate at the local end cannot be correctly measured. This parameter takes effect only when TWAMP is deployed in light architecture.
Local IP Address	**TWAMPCLIENT**.***LOCALIP***	N/A

Parameter Name	Parameter ID	Setting Notes
Peer IP Address	**TWAMPCLIENT**.***PEERIP***	N/A
Peer TCP Port No.	**TWAMPCLIENT**.***PEERPORT***	Indicates the peer TCP port number of a TWAMP client. This parameter is valid only when TWAMP is deployed in full architecture. The default value of this parameter is **862**, and this parameter value must be negotiated with the TWAMP responder.
Client Index	**TWAMPCLIENT**.***CLIENTID***	N/A
DSCP	**TWAMPCLIENT**.***DSCP***	This parameter is valid only in full architecture. The default value is **46**.
DSCP	**TWAMPSENDER**.***DSCP***	It is recommended that you set this parameter to the priority of the service packets for which users show concern. The default value is **0**.
Packet Size Type	**TWAMPSENDER**.***PKTSIZETYPE***	The default value **FIXED** is recommended.
Packet Size	**TWAMPSENDER**.***PKTSIZE***	Retain the default value **128**.
Packet Send Interval Type	**TWAMPSENDER**.***PKTINTTYPE***	Retain the default value **FIXED**.
Packet Send Interval	**TWAMPSENDER**.***PKTINT***	Indicates the interval at which packets are sent. The default value is **1000**.
Local UDP Port No.	**TWAMPSENDER**.***LOCALUDPPORT***	This parameter is valid only when TWAMP is deployed in light architecture. The value of this parameter must be negotiated with the TWAMP responder.
Peer UDP Port No.	**TWAMPSENDER**.***PEERUDPPORT***	This parameter is valid only when TWAMP is deployed in light architecture. The value of this parameter must be negotiated with the TWAMP responder.

Table 4-11 Data preparation for the base station serving as the TWAMP Controller

Parameter Name	Parameter ID	Setting Notes
TWAMP Architecture	**TWAMPCLIENT**.***TWAMPARCH (LTE eNodeB, 5G gNodeB)***	Set this parameter to a value consistent with that of the TWAMP responder.
Light Architecture Mode	**TWAMPSENDER**.***LIGHTARCHITECTUREMODE (LTE eNodeB, 5G gNodeB)***	Set this parameter to a value consistent with that of the TWAMP responder. Otherwise, the counters related to the packet loss rate at the local end cannot be correctly measured. This parameter takes effect only when TWAMP is deployed in light architecture.
IP Version	**TWAMPCLIENT**.***IPVERSION (LTE eNodeB, 5G gNodeB)***	N/A
Local IP Address	**TWAMPCLIENT**.***LOCALIP (LTE eNodeB, 5G gNodeB)***	N/A
Peer IP Address	**TWAMPCLIENT**.***PEERIP (LTE eNodeB, 5G gNodeB)***	N/A
Local IPv6 Address	**TWAMPCLIENT**.***LOCALIPV6 (LTE eNodeB, 5G gNodeB)***	N/A
Peer IPv6 Address	**TWAMPCLIENT**.***PEERIPV6 (LTE eNodeB, 5G gNodeB)***	N/A
Peer TCP Port No.	**TWAMPCLIENT**.***PEERPORT (LTE eNodeB, 5G gNodeB)***	Indicates the peer TCP port number of a TWAMP client. This parameter is valid only when TWAMP is deployed in full architecture. The default value of this parameter is **862**, and this parameter value must be negotiated with the TWAMP responder.
Client Index	**TWAMPCLIENT**.***CLIENTID (LTE eNodeB, 5G gNodeB)***	N/A
VRF Index	**TWAMPCLIENT**.***VRFINDEX (LTE eNodeB, 5G gNodeB)***	Indicates the virtual routing domain to which the local IP address or local IPv6 address belongs. The default value is **0**.

Parameter Name	Parameter ID	Setting Notes
DSCP	**TWAMPCLIENT**.***DSCP (LTE eNodeB, 5G gNodeB)***	This parameter is valid only in full architecture. The default value is **46**.
DSCP	**TWAMPSENDER**.***DSCP (LTE eNodeB, 5G gNodeB)***	It is recommended that you set this parameter to the priority of the service packets for which users show concern. The default value is **0**.
Individual Session Control Switch	**TWAMPCLIENT**.***INDIVSESSCTRLSW (LTE eNodeB, 5G gNodeB)***	This parameter is valid only in full architecture. The default value is **DISABLE**. The individual session control function takes effect only when the Controller and Responder both comply with RFC 5938.
Packet Size Type	**TWAMPSENDER**.***PKTSIZETYPE (LTE eNodeB, 5G gNodeB)***	The default value **FIXED** is recommended.
Packet Size	**TWAMPSENDER**.***PKTSIZE (LTE eNodeB, 5G gNodeB)***	Retain the default value **128**.
Packet Send Interval Type	**TWAMPSENDER**.***PKTINTTYPE (LTE eNodeB, 5G gNodeB)***	Retain the default value **FIXED**.
Packet Send Interval	**TWAMPSENDER**.***PKTINT (LTE eNodeB, 5G gNodeB)***	Indicates the interval at which packets are sent. The default value is **1000**.
Local UDP Port No.	**TWAMPSENDER**.***LOCALUDPPORT (LTE eNodeB, 5G gNodeB)***	This parameter is valid only when TWAMP is deployed in light architecture. The value of this parameter must be negotiated with the TWAMP responder.
Peer UDP Port No.	**TWAMPSENDER**.***PEERUDPPORT (LTE eNodeB, 5G gNodeB)***	This parameter is valid only when TWAMP is deployed in light architecture. The value of this parameter must be negotiated with the TWAMP responder.

NOTE

The local end actively transmits packets when it functions as the TWAMP Controller. The bandwidth consumed by the transmitted packets can be set using the ***PKTSIZE (LTE eNodeB, 5G gNodeB)*** and ***PktInt (LTE eNodeB, 5G gNodeB)*** parameters.

- Services packets are simulated in TWAMP tests, which occupy some bandwidth. To prevent services from being affected, it is recommended that you enable IP Active Performance Measurement only when you are familiar with this feature and TWAMP. To minimize risks and negative impacts on services, the ***PKTSIZE (LTE eNodeB, 5G gNodeB)*** and ***PKTINT (LTE eNodeB, 5G gNodeB)*** parameters are set to their default values **128** and **1000**, respectively. In this case, the bandwidth consumed by the transmitted packets is only 1 kbit/s.
- TWAMP testing uses packet injection and the test accuracy is related to the packet transmission rate. The greater the packet transmission rate, the higher the accuracy. You can modify the ***PKTINT (LTE eNodeB, 5G gNodeB)*** parameter to increase the packet transmission rate for higher test accuracy if there is sufficient network bandwidths.

Table 4-12 Data preparation for the base station controller BSC6900/BSC6910 serving as the TWAMP Responder

Parameter Name	Parameter ID	Setting Notes
Local IP Address	**TWAMPRESPONDER**.***LOCALIP***	N/A
Local TCP Port No.	**TWAMPRESPONDER**.***LOCALPORT***	N/A
Responder Index	**TWAMPRESPONDER**.***RESPONDERID***	N/A
DSCP for TCP Packets	**TWAMPRESPONDER**.***DSCP***	The default value is **46**.
Negotiation Wait Time	**TWAMPRESPONDER**.***SERVWAIT***	The default value, **900**, defined by the protocol is recommended.
Measurement Wait Time	**TWAMPRESPONDER**.***REFWAIT***	The default value, **900**, defined by the protocol is recommended.

Table 4-13 Data preparation for the base station controller BSC6960 serving as the TWAMP Responder

Parameter Name	Parameter ID	Setting Notes
Local IP Address	**TWAMPRESPONDER**.***LOCALIP***	N/A

Parameter Name	Parameter ID	Setting Notes
TWAMP Architecture	**TWAMPRESPONDER**.***TWAMPARCH***	Set this parameter to a value consistent with that on the Controller end.
Light Architecture Local UDP Port No.	**TWAMPRESPONDER**.***LIGHTLOCALUDPPORT***	This parameter is valid only in light architecture. The value of this parameter must be negotiated with the TWAMP Session-Sender.
Light Architecture Mode	**TWAMPRESPONDER**.***LIGHTARCHITECTUREMODE***	Set this parameter to a value consistent with that of the TWAMP Session-Sender. Otherwise, the counters for the TWAMP Controller cannot be correctly measured. This parameter takes effect only when TWAMP is deployed in light architecture.
Local TCP Port No.	**TWAMPRESPONDER**.***LOCALPORT***	This parameter is valid only in full architecture. The value of this parameter must be negotiated with the TWAMP Controller.
Responder Index	**TWAMPRESPONDER**.***RESPONDERID***	N/A
DSCP for TCP Packets	**TWAMPRESPONDER**.***DSCP***	The default value is **46**.
Negotiation Wait Time	**TWAMPRESPONDER**.***SERVWAIT***	This parameter is valid only in full architecture. The protocol-defined default value **900** is recommended.
Measurement Wait Time	**TWAMPRESPONDER**.***REFWAIT***	The protocol-defined default value **900** is recommended.

Table 4-14 Data preparation for the base station serving as the TWAMP Responder

Parameter Name	Parameter ID	Setting Notes
IP Version	**TWAMPRESPONDER**.***IPVERSION (LTE eNodeB, 5G gNodeB)***	N/A

Parameter Name	Parameter ID	Setting Notes
Local IP Address	**TWAMPRESPONDER**.***LOCALIP (LTE eNodeB, 5G gNodeB)***	N/A
Local IPv6 Address	**TWAMPRESPONDER**.***LOCALIPV6 (LTE eNodeB, 5G gNodeB)***	N/A
TWAMP Architecture	**TWAMPRESPONDER**.**TWAMPARCH (*LTE eNodeB, 5G gNodeB)***	Set this parameter to a value consistent with that on the Controller end.
Light Architecture Local UDP Port No.	**TWAMPRESPONDER**.**LIGHTLOCALUDPPORT (*LTE eNodeB, 5G gNodeB)***	This parameter is valid only in light architecture. The value of this parameter must be negotiated with the TWAMP Session-Sender.
Light Architecture Mode	**TWAMPRESPONDER**.***LIGHTARCHITECTUREMODE (LTE eNodeB, 5G gNodeB)***	Set this parameter to a value consistent with that of the TWAMP Session-Sender. Otherwise, the counters for the TWAMP Controller cannot be correctly measured. This parameter takes effect only when TWAMP is deployed in light architecture.
Local TCP Port No.	**TWAMPRESPONDER**.**LOCALPORT (*LTE eNodeB, 5G gNodeB)***	This parameter is valid only in full architecture. The value of this parameter must be negotiated with the TWAMP Controller.
Responder Index	**TWAMPRESPONDER**.***RESPONDERID (LTE eNodeB, 5G gNodeB)***	N/A
VRF Index	**TWAMPRESPONDER**.**VRFINDEX (*LTE eNodeB, 5G gNodeB)***	Indicates the virtual routing domain to which the local IP address or local IPv6 address belongs. The default value is **0**.
DSCP for TCP Packets	**TWAMPRESPONDER**.**DSCP (*LTE eNodeB, 5G gNodeB)***	The default value is **46**.
Negotiation Wait Time	**TWAMPRESPONDER**.**SERVWAIT (*LTE eNodeB, 5G gNodeB)***	This parameter is valid only in full architecture. The protocol-defined default value **900** is recommended.

Parameter Name	Parameter ID	Setting Notes
Measurement Wait Time	**TWAMPRESPONDER.REFWAIT** ***(LTE eNodeB, 5G gNodeB)***	The protocol-defined default value **900** is recommended.

Table 4-15 TWAMP communication matrix requirements for base stations

Protocol	Source Device	Source Port	Destination Device	Destination Port	Port Description
TCP	Base station (TWAMP Control-Client)	49152–65535	TWAMP server	1–65535	Used by a base station serving as the TWAMP Control-Client to send TWAMP control packets to the TWAMP server. The destination port on the base station side must be set to the listening port on the TWAMP server side. The protocol-defined default port number is 862.
	TWAMP server	1–65535	Base station (TWAMP Control-Client)	1024–65535	Used by a base station serving as the TWAMP Control-Client to receive TWAMP server's responses to TWAMP control packets.
	TWAMP Control-Client	Specified by the client	Base station (TWAMP server)	862, 1024–65535	Used by a base station serving as the TWAMP server to receive control packets from the TWAMP Control-Client. The port listened by the base station is configured by users. The default port number is 862.
	Base station (TWAMP server)	862, 1024–65535	TWAMP Control-Client	Specified by the client	Used by a base station serving as the TWAMP server to send control packets to the TWAMP Control-Client.
UDP	Base station (TWAMP Session-Sender)	64695–64710	TWAMP Session-Reflector	Specified by the server	Used by a base station serving as the TWAMP Session-Sender to send test packets to the TWAMP Reflector.

Protocol	Source Device	Source Port	Destination Device	Destination Port	Port Description
	TWAMP Session-Reflector	Specified by the server	Base station (TWAMP Session-Sender)	64695–64710	Used by a base station serving as the TWAMP Session-Sender to receive test packets from the TWAMP Reflector.
	TWAMP Session-Sender	Specified by the client	Base station (TWAMP Session-Reflector)	862, 64679–64694	Used by a base station serving as the TWAMP Reflector to send test packets to the TWAMP Session-Sender.
	Base station (TWAMP Session-Reflector)	862, 64679–64694	TWAMP Session-Sender	Specified by the client	Used by a base station serving as the TWAMP Reflector to receive test packets from the TWAMP Session-Sender.

Table 4-16 TWAMP communication matrix requirements for base station controllers

Protocol	Source Device	Source Port	Destination Device	Destination Port	Port Description
TCP	Base station controller (TWAMP Control-Client)	5201–6225 (BSC6900/BSC6910) 49152–65535 (BSC6960)	Base station (TWAMP server)	862, 1024–65535	Used by a base station controller serving as the TWAMP Control-Client to send TWAMP control packets to a base station, which serves as the TWAMP server. The destination port on the base station controller side must be set to the port listened by the TWAMP server side. The protocol-defined default port number is 862.

Protocol	Source Device	Source Port	Destination Device	Destination Port	Port Description
	Base station (TWAMP server)	862, 1024–65535	Base station controller (TWAMP Control-Client)	5201–6225 (BSC6900/ BSC6910) 49152–65535 (BSC6960)	Used by a base station controller serving as the TWAMP Control-Client to receive responses to TWAMP control packets from a base station, which functions as the TWAMP server
	Base station (TWAMP Control-Client)	49152–65535	Base station controller (TWAMP server)	862, 1024–65535	Used by a base station controller serving as the TWAMP server to receive control packets from a base station, which serves as the TWAMP Control-Client. The port listened by the base station controller is configured by users. The default port number is 862.
	Base station controller (TWAMP server)	862, 1024–65535	Base station (TWAMP Control-Client)	49152–65535	Used by a base station controller serving as the TWAMP server to send control packets to a base station, which serves as the TWAMP Control-Client
UDP	Base station controller (TWAMP Session-Sender)	64968–64983 (BSC6900/ BSC6910) 64695–64710 (BSC6960)	Base station (TWAMP Session-Reflector)	862, 64679–64694	Used by a base station controller serving as the TWAMP Session-Sender to send test packets to a base station, which serves as the TWAMP Reflector
	Base station (TWAMP Session-Reflector)	862, 64679–64694	Base station controller (TWAMP Session-Sender)	64968–64983 (BSC6900/ BSC6910) 64695–64710 (BSC6960)	Used by a base station controller serving as the TWAMP Session-Sender to receive test packets from a base station, which serves as the TWAMP Reflector

Protocol	Source Device	Source Port	Destination Device	Destination Port	Port Description
	Base station (TWAMP Session-Sender)	64695–64710	Base station controller (TWAMP Session-Reflector)	862, 64984–64999 (BSC6900/BSC6910) 862, 64679–64694 (BSC6960)	Used by a base station controller serving as the TWAMP Reflector to send test packets to a base station, which serves as the TWAMP Session-Sender
	Base station controller (TWAMP Session-Reflector)	862, 64984–64999 (BSC6900/BSC6910) 862, 64679–64694 (BSC6960)	Base station (TWAMP Session-Sender)	64695–64710	Used by a base station controller serving as the TWAMP Reflector to receive test packets from the TWAMP Session-Sender

- When serving as the TWAMP Controller in full architecture, the local end sends the TWAMP Responder a Request-TW-Session message with the Receiver Port number allocated to the peer end being a fixed value during the negotiation. If the TWAMP Responder does not accept the Receiver Port number sent by the local end, it must reply with an Accept-Session message containing an appropriate UDP port number. Otherwise, the negotiation may fail.
- When the local end serves as the TWAMP Responder in light architecture, **TWAMPRESPONDER**.***LIGHTLOCALUDPPORT (LTE eNodeB, 5G gNodeB)*** must be manually configured at the local end. The value of this parameter ranges from **64679** to **64694**, as described in **Table 4-15**. In addition, the Sender UDP port number manually configured at the peer end must be identical with local UDP port number. Otherwise, the interconnection may fail.
- When the local end serves as the TWAMP Controller and the light architecture is used, the **TWAMPSENDER**.***LOCALUDPPORT (LTE eNodeB, 5G gNodeB)*** and **TWAMPSENDER**.***PEERUDPPORT (LTE eNodeB, 5G gNodeB)*** parameters must be manually configured at the local end. The **TWAMPSENDER**.***PEERUDPPORT (LTE eNodeB, 5G gNodeB)*** parameter must be set to a value same as that of the Responder UDP port number manually configured at the peer end. Otherwise, the interconnection may fail.

4.5.1.2 Using MML Commands

Before using MML commands, refer to **4.4 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples for TWAMP Full Architecture

The base station controller serves as the TWAMP Controller.

```
//Configuring the base station controller as the TWAMP Control-Client
ADD TWAMPCLIENT: CLIENTID=0, LOCALIP="192.168.11.110", PEERIP="192.168.22.220", PEERPORT=862;
//Configuring the base station controller as the TWAMP Session-Sender
ADD TWAMPSENDER: CLIENTID=0, SENDERID=0, PKTSIZETYPE=FIXED, PKTINTTYPE=FIXED;
```

The base station serves as the TWAMP Controller.

```
//Configuring the base station as the TWAMP Client, and setting the IP version to IPv4
ADD TWAMPCLIENT: CLIENTID=0, IPVERSION=IPv4,LOCALIP="192.168.11.110", PEERIP="192.168.22.220",
TWAMPARCH=FULL, PEERPORT=862, INDIVSESSCTRLSW = DISABLE;
//Configuring the base station as the TWAMP Client, and setting the IP version to IPv6
ADD TWAMPCLIENT: CLIENTID=0, IPVERSION=IPv6,LOCALIPV6="2015:1960::123", PEERIPV6="2015:240::11",
TWAMPARCH=FULL, PEERPORT=862, INDIVSESSCTRLSW = DISABLE;
//Configuring the base station as the TWAMP Session-Sender
ADD TWAMPSENDER: CLIENTID=0, SENDERID=0,TWAMPARCH=FULL, PKTSIZETYPE=FIXED,
PKTINTTYPE=FIXED;
```

NOTE

- A modification to the **TWAMPCLIENT** or **TWAMPSENDER** MO in full architecture results in a TWAMP re-negotiation, which takes approximately two minutes.
- Removal of some TWAMPSENDER MOs in full architecture results in a TWAMP re-negotiation, which takes approximately two minutes.

The base station controller serves as the TWAMP Responder.

```
//Configuring the base station controller as the TWAMP Responder
ADD TWAMPRESPONDER: RESPONDERID=0, LOCALIP="192.168.22.220", LOCALPORT=862;
```

The base station serves as the TWAMP Responder.

```
//Configuring the base station as the TWAMP Responder, and setting the IP version to IPv4
ADD TWAMPRESPONDER: RESPONDERID=0, IPVERSION=IPv4, LOCALIP="192.168.22.220",
TWAMPARCH=FULL, LOCALPORT=862;
//Configuring the base station as the TWAMP Responder, and setting the IP version to IPv6
ADD TWAMPRESPONDER: RESPONDERID=0, IPVERSION=IPv6, LOCALIPV6="2015:240::11",
TWAMPARCH=FULL, LOCALPORT=862;
```

NOTE

A modification to the **TWAMPRESPONDER** MO in full architecture results in a TWAMP re-negotiation.

Activation Command Examples for TWAMP Light Architecture

In light architecture, you are advised to activate the TWAMP Responder function prior to the TWAMP Controller function.

The base station serves as the TWAMP Responder.

```
//Configuring the base station using IPv6 transmission as the TWAMP Responder in stateful mode of light architecture
ADD TWAMPRESPONDER: RESPONDERID=0, IPVERSION=IPv6, LOCALIPV6="2015:240::11", TWAMPARCH=LIGHT, LIGHTLOCALUDPPORT=64679;
//(Optional) Configuring the base station using IPv6 transmission as the TWAMP Responder in stateless mode of light architecture
ADD TWAMPRESPONDER: RESPONDERID=1, IPVERSION=IPv6, LOCALIPV6="2015:240::11", TWAMPARCH=LIGHT, LIGHTLOCALUDPPORT=64680, LIGHTARCHITECTUREMODE=STATELESS;
//Configuring the base station using IPv4 transmission as the TWAMP Responder in stateful mode of light architecture
ADD TWAMPRESPONDER: RESPONDERID=2, IPVERSION=IPv4, LOCALIP="192.168.22.220", TWAMPARCH=LIGHT, LIGHTLOCALUDPPORT=64679;
//(Optional) Configuring the base station using IPv4 transmission as the TWAMP Responder in stateless mode of light architecture
ADD TWAMPRESPONDER: RESPONDERID=3, IPVERSION=IPv4,LOCALIP="192.168.22.220", TWAMPARCH=LIGHT, LIGHTLOCALUDPPORT=64680, LIGHTARCHITECTUREMODE=STATELESS;
```

The base station serves as the TWAMP Controller.

```
//Configuring the base station as the TWAMP Client, and setting the IP version to IPv6
ADD TWAMPCLIENT: CLIENTID=0, IPVERSION=IPv6,LOCALIPV6="2015:1960::123", PEERIPV6="2015:240::11", TWAMPARCH=LIGHT;
//Configuring the base station as the TWAMP Session-Sender in stateful mode of light architecture
ADD TWAMPSENDER: CLIENTID=0, SENDERID=0, TWAMPARCH=LIGHT, LOCALUDPPORT=64695, PEERUDPPORT=64679, DSCP=18, PKTSIZETYPE=FIXED, PKTINTTYPE=FIXED;
//(Optional) Configuring the base station as the TWAMP Session-Sender in stateless mode of light architecture
ADD TWAMPSENDER: CLIENTID=0, SENDERID=1, TWAMPARCH=LIGHT, LIGHTARCHITECTUREMODE=STATELESS, LOCALUDPPORT=64696, PEERUDPPORT=64680, DSCP=46, PKTSIZETYPE=FIXED, PKTINTTYPE=FIXED;
//Configuring the base station as the TWAMP Client, and setting the IP version to IPv4
ADD TWAMPCLIENT: CLIENTID=0, IPVERSION=IPv4, LOCALIP="192.168.11.110", PEERIP="192.168.22.220", TWAMPARCH=LIGHT;
//Configuring the base station as the TWAMP Session-Sender in stateful mode of light architecture
ADD TWAMPSENDER: CLIENTID=0, SENDERID=0, TWAMPARCH=LIGHT, LOCALUDPPORT=64695, PEERUDPPORT=64679, DSCP=18, PKTSIZETYPE=FIXED, PKTINTTYPE=FIXED;
//(Optional) Configuring the base station as the TWAMP Session-Sender in stateless mode of light architecture
ADD TWAMPSENDER: CLIENTID=0, SENDERID=1, TWAMPARCH=LIGHT, LIGHTARCHITECTUREMODE=STATELESS, LOCALUDPPORT=64696, PEERUDPPORT=64680, DSCP=46, PKTSIZETYPE=FIXED, PKTINTTYPE=FIXED;
```

NOTE

To ensure the accuracy of measurement results in light architecture:

- You are not advised to change the mode of the light architecture during ongoing TWAMP measurement. If a mode change is required, deactivate the devices at both ends, and activate them in the activation sequence after the change.
- If the TWAMP Controller or TWAMP Responder function needs to be temporarily interrupted during ongoing TWAMP measurement, deactivate the TWAMP Responder or TWAMP Controller function as well.

The sequence number-based loop breaking mechanism is implemented on the base station serving as the TWAMP Responder in stateless mode of TWAMP light deployment, and the detection window size is 10. Specifically, the base station does not respond when a received sequence number is the same as any sequence number in the previous ten receptions. The TWAMP Session-Sender must comply with the standard, and the sequence number in a session is incremented by one in ascending order within a certain range.

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Removing the TWAMP Session-Sender
RMV TWAMPSENDER: CLIENTID=0, SENDERID=0;
//Removing the TWAMP Control-Client
RMV TWAMPCLIENT: CLIENTID=0;
```

NOTE

You must remove all the Session-Senders corresponding to a Control-Client before removing the Control-Client.

```
//Deactivating the TWAMP Responder function
RMV TWAMPRESPONDER: RESPONDERID=0;
```

4.5.2 Activation Verification

Use one of the following methods to verify whether the TWAMP functions have been successfully activated.

Table 4-17 Activation verification methods

NE	Using MML Commands	Using Performance Statistics	Using the Real-Time Performance Monitoring Task
Base station	Supported. The performance measurement results are valid only when the NE serves as the TWAMP sender. The UMPT or UMDU supports the measurement of microsecond-level and millisecond-level RTT and delay variation results. Other boards support only the millisecond-level measurement results.	Supported. This method is applicable only when the NE serves as the TWAMP sender. The UMPT or UMDU supports the measurement of microsecond-level and millisecond-level RTT and delay variation results. Other boards support only the millisecond-level measurement results.	Supported. This method is applicable only when the NE serves as the TWAMP sender. The UMPT or UMDU supports the measurement of microsecond-level and millisecond-level RTT and delay variation results. Other boards support only the millisecond-level measurement results.

NE	Using MML Commands	Using Performance Statistics	Using the Real-Time Performance Monitoring Task
Base station controller	Supported. The performance measurement results are valid only when the NE serves as the TWAMP sender.	Supported. This method is applicable only when the NE serves as the TWAMP sender.	Not supported.

NOTE

When a base station and a base station controller work as the TWAMP sender and TWAMP responder, respectively, only two-way delay measurement is supported. One-way delay measurement is not supported.

Using MML Commands

Assume that the NE complies with RFC 5357. The following describes the activation verification for the TWAMP full architecture.

If an NE serves as the TWAMP Controller in full architecture, perform the following operations for activation observation:

Step 1 Query the local Control-Client status by running the **DSP TWAMPCLIENT** command 2 minutes after the TWAMP Controller function is enabled. For a base station controller (BSC6910/BSC6900), if the value of **Negotiation Status** is **Negotiation succeeded**, TWAMP negotiation on the CP is successful. For a base station or base station controller (BSC6960), if the value of **Negotiation Status** is **TCP_LINK_UP**, TWAMP negotiation on the CP is successful.

Step 2 Query the local Session-Sender status by running the **DSP TWAMPSENDER** command. For a base station controller (BSC6910/BSC6900), if the value of **Negotiation Status** is **Negotiation succeeded**, the TWAMP test session has been negotiated successfully. For a base station or base station controller (BSC6960), if the value of **Negotiation Status** is **SESSION_UP**, the TWAMP test session has been negotiated successfully.

----End

If an NE serves as the TWAMP Responder in full architecture, perform the following operation for activation observation:

Query the local Responder status by running the **DSP TWAMPRESPONDER** command. For a base station controller (BSC6910/BSC6900), if the value of **Negotiation Status** is **Session succeeded**, the TWAMP test session has been negotiated successfully. For a base station/base station controller (BSC6960), if the value of **Negotiation Status** is **SESSION_UP**, the TWAMP test session has been negotiated successfully.

The following describes the activation observation for the TWAMP light architecture.

If a base station or base station controller (BSC6960) serves as the TWAMP Controller in light architecture, perform the following operation for activation verification: Query the local Sender status by running the **DSP TWAMPSENDER** command. If the TWAMP session properly sends and receives test packets, the TWAMP session has been successfully established.

Assume that the NE complies with RFC 5938. The following describes the activation verification for the TWAMP full architecture in individual session control mode.

If an NE serves as the TWAMP Controller in full architecture, perform the following operations for activation verification:

Step 1 Query the local Control-Client status by running the **DSP TWAMPCLIENT** command 2 minutes after the TWAMP Controller function is enabled and the **Individual Session Control Switch** parameter is set to **ENABLE**. For a base station or base station controller (BSC6960), if the value of **Individual Session Control Switch State** is **ENABLE** in the command output, the individual session control function has taken effect. If the value is **DISABLE**, the peer end does not support this function, leading to a failure in the negotiation with the peer end. As a result, the function does not take effect.

Step 2 Query the local Session-Sender status by running the **DSP TWAMPSENDER** command. For a base station or base station controller (BSC6960), if the value of **Negotiation Status** is **SESSION_UP**, the TWAMP test session has been negotiated successfully.

----End

If an NE serves as the TWAMP Responder in full architecture, perform the following operations for activation observation:

Query the local Responder status by running the **DSP TWAMPRESPONDER** command. For a base station or base station controller (BSC6960), if the value of **Individual Session Control Switch State** is **ENABLE** in the command output, the individual session control function has taken effect. If the value is **DISABLE**, the peer end does not support this function, leading to a failure in the negotiation with the peer end. As a result, the function does not take effect.

If the base station or base station controller (BSC6960) serves as the TWAMP Responder in light architecture, perform the following operation for activation observation: Query the local Responder status by running the **DSP TWAMPRESPONDER** command. If the query results are consistent with the settings of the TWAMP Controller, and the TWAMP test session properly sends and receives test packets, the TWAMP test session has been successfully established.

Using Performance Statistics

Collect values of TWAMP performance counters for each NE. If any of the counter values is not null, the IP Active Performance Measurement feature has been successfully activated. **Table 4-18Table 4-19** and **Table 4-20** provide TWAMP performance counters on the base station controller side and base station side, respectively.

Table 4-18 TWAMP performance counters on the base station controller side (when the BSC6900/BSC6910 is used)

Counter ID	Counter Name	Counter Description
73443310	**VS.TWAMP.Forward.DropRates.Mean**	Average Forward Packet Loss Rates for TWAMP Measurement
73443304	**VS.TWAMP.Forward.DropRates.Max**	Peak Forward Packet Loss Rates for TWAMP Measurement
73443305	**VS.TWAMP.Backward.DropRates.Mean**	Average Backward Packet Loss Rates for TWAMP Measurement
73443306	**VS.TWAMP.Backward.DropRates.Max**	Peak Backward Packet Loss Rates for TWAMP Measurement
73426982	**VS.TWAMP.RttDelay.Min**	Minimum RTT for TWAMP Measurement
73443307	**VS.TWAMP.RttDelay.Mean**	Average RTT for TWAMP Measurement
73426983	**VS.TWAMP.RttDelay.Max**	Maximum RTT for TWAMP Measurement
73426985	**VS.TWAMP.Forward.Jitter.Min**	Minimum Forward Delay Jitters for TWAMP Measurement
73443308	**VS.TWAMP.Forward.Jitter.Mean**	Average Forward Delay Jitters for TWAMP Measurement
73426986	**VS.TWAMP.Forward.Jitter.Max**	Maximum Forward Delay Jitters for TWAMP Measurement
73426989	**VS.TWAMP.Backward.Jitter.Min**	Minimum Backward Delay Jitters for TWAMP Measurement
73443309	**VS.TWAMP.Backward.Jitter.Mean**	Average Backward Delay Jitters for TWAMP Measurement
73426990	**VS.TWAMP.Backward.Jitter.Max**	Maximum Backward Delay Jitters for TWAMP Measurement

Table 4-19 TWAMP performance counters on the base station controller side (when the BSC6960 is used)

Counter ID	Counter Name	Counter Description
1542455996	**VS.TWAMP.Forward.DropMeans**	Average Forward Packet Loss Rates for TWAMP Measurement

Counter ID	Counter Name	Counter Description
1542455997	**VS.TWAMP.Forward.Peak.DropRates**	Peak Forward Packet Loss Rates for TWAMP Measurement
1542455998	**VS.TWAMP.Backward.DropMeans**	Average Backward Packet Loss Rates for TWAMP Measurement
1542455999	**VS.TWAMP.Backward.Peak.DropRates**	Peak Backward Packet Loss Rates for TWAMP Measurement
1542456000	**VS.TWAMP.MinRttDelay**	Minimum RTT for TWAMP Measurement
1542456001	**VS.TWAMP.Rtt.Means**	Average RTT for TWAMP Measurement
1542456002	**VS.TWAMP.MaxRttDelay**	Maximum RTT for TWAMP Measurement
1542456003	**VS.TWAMP.Forward.MinJitter**	Minimum Forward Delay Jitters for TWAMP Measurement
1542456004	**VS.TWAMP.Forward.Jitter.Means**	Average Forward Delay Jitters for TWAMP Measurement
1542456005	**VS.TWAMP.Forward.MaxJitter**	Maximum Forward Delay Jitters for TWAMP Measurement
1542456006	**VS.TWAMP.Backward.MinJitter**	Minimum Backward Delay Jitters for TWAMP Measurement
1542456007	**VS.TWAMP.Backward.Jitter.Means**	Average Backward Delay Jitters for TWAMP Measurement
1542456008	**VS.TWAMP.Backward.MaxJitter**	Maximum Backward Delay Jitters for TWAMP Measurement
1542460412	**VS.TWAMP.Sender.TxPackets**	Number of packets transmitted by the sender on the BSTWAMP
1542460413	**VS.TWAMP.Reflector.TxPackets**	Number of packets transmitted by the reflector on the BSTWAMP
1542460414	**VS.TWAMP.Sender.RxPackets**	Number of packets received by the sender on the BSTWAMP
1542460725	**VS.TWAMP.RoundTrip.DropMeans**	Two-way average packet loss rate on the BSTWAMP
1542460726	**VS.TWAMP.RoundTrip.Peak.DropRates**	Two-way peak packet loss rate on the BSTWAMP
1542460755	**VS.TWAMP.HighLevel.MinRttDelay**	Minimum high-precision RTT on the BSTWAMP

Counter ID	Counter Name	Counter Description
1542460756	**VS.TWAMP.HighLevel.Rtt.Means**	Average high-precision RTT on the BSTWAMP
1542460757	**VS.TWAMP.HighLevel.MaxRttDelay**	Maximum high-precision RTT on the BSTWAMP
1542460758	**VS.TWAMP.HighLevel.Forward.MinJitter**	Minimum high-precision forward jitter on the BSTWAMP
1542460759	**VS.TWAMP.HighLevel.Forward.Jitter.Means**	Average high-precision forward jitter on the BSTWAMP
1542460760	**VS.TWAMP.HighLevel.Forward.MaxJitter**	Maximum high-precision forward jitter on the BSTWAMP
1542460761	**VS.TWAMP.HighLevel.Backward.MinJitter**	Minimum high-precision backward jitter on the BSTWAMP
1542460762	**VS.TWAMP.HighLevel.Backward.Jitter.Means**	Average high-precision backward jitter on the BSTWAMP
1542460763	**VS.TWAMP.HighLevel.Backward.MaxJitter**	Maximum high-precision backward jitter on the BSTWAMP

Table 4-20 TWAMP performance counters on the base station side

Counter ID	Counter Name	Counter Description
1542455996	**VS.BSTWAMP.Forward.DropMeans (LTE eNodeB, 5G gNodeB)**	Average forward packet loss rate on the BSTWAMP
1542455997	**VS.BSTWAMP.Forward.Peak.DropRates (LTE eNodeB, 5G gNodeB)**	Peak forward packet loss rate on the BSTWAMP
1542455998	**VS.BSTWAMP.Backward.DropMeans (LTE eNodeB, 5G gNodeB)**	Average backward packet loss rate on the BSTWAMP
1542455999	**VS.BSTWAMP.Backward.Peak.DropRates (LTE eNodeB, 5G gNodeB)**	Peak backward packet loss rate on the BSTWAMP
1542460725	**VS.BSTWAMP.RoundTrip.DropMeans (LTE eNodeB, 5G gNodeB)**	Two-way average packet loss rate on the BSTWAMP

Counter ID	Counter Name	Counter Description
1542460726	**VS.BSTWAMP.RoundTrip.Peak.DropRates (LTE eNodeB, 5G gNodeB)**	Two-way peak packet loss rate on the BSTWAMP
1542456000	**VS.BSTWAMP.MinRttDelay (LTE eNodeB, 5G gNodeB)**	Minimum RTT on the BSTWAMP
1542456001	**VS.BSTWAMP.Rtt.Means (LTE eNodeB, 5G gNodeB)**	Average RTT on the BSTWAMP
1542456002	**VS.BSTWAMP.MaxRttDelay (LTE eNodeB, 5G gNodeB)**	Maximum RTT on the BSTWAMP
1542460755	**VS.BSTWAMP.HighLevel.MinRttDelay (LTE eNodeB, 5G gNodeB)**	Minimum high-precision RTT on the BSTWAMP
1542460756	**VS.BSTWAMP.HighLevel.Rtt.Means (LTE eNodeB, 5G gNodeB)**	Average high-precision RTT on the BSTWAMP
1542460757	**VS.BSTWAMP.HighLevel.MaxRttDelay (LTE eNodeB, 5G gNodeB)**	Maximum high-precision RTT on the BSTWAMP
1542456003	**VS.BSTWAMP.Forward.MinJitter (LTE eNodeB, 5G gNodeB)**	Minimum forward jitter on the BSTWAMP
1542456004	**VS.BSTWAMP.Forward.Jitter.Means (LTE eNodeB, 5G gNodeB)**	Average Forward Delay Jitters for TWAMP Measurement
1542456005	**VS.BSTWAMP.Forward.MaxJitter (LTE eNodeB, 5G gNodeB)**	Maximum forward jitter on the BSTWAMP
1542460758	**VS.BSTWAMP.HighLevel.Forward.MinJitter (LTE eNodeB, 5G gNodeB)**	Minimum high-precision forward jitter on the BSTWAMP
1542460759	**VS.BSTWAMP.HighLevel.Forward.Jitter.Means (LTE eNodeB, 5G gNodeB)**	Average high-precision forward jitter on the BSTWAMP

Counter ID	Counter Name	Counter Description
1542460760	**VS.BSTWAMP.HighLevel.Forward.MaxJitter (LTE eNodeB, 5G gNodeB)**	Maximum high-precision forward jitter on the BSTWAMP
1542456006	**VS.BSTWAMP.Backward.MinJitter (LTE eNodeB, 5G gNodeB)**	Minimum backward jitter on the BSTWAMP
1542456007	**VS.BSTWAMP.Backward.Jitter.Means (LTE eNodeB, 5G gNodeB)**	Average backward jitter on the BSTWAMP
1542456008	**VS.BSTWAMP.Backward.MaxJitter (LTE eNodeB, 5G gNodeB)**	Maximum Backward Delay Jitters for TWAMP Measurement
1542460761	**VS.BSTWAMP.HighLevel.Backward.MinJitter (LTE eNodeB, 5G gNodeB)**	Minimum high-precision backward jitter on the BSTWAMP
1542460762	**VS.BSTWAMP.HighLevel.Backward.Jitter.Means (LTE eNodeB, 5G gNodeB)**	Average high-precision backward jitter on the BSTWAMP
1542460763	**VS.BSTWAMP.HighLevel.Backward.MaxJitter (LTE eNodeB, 5G gNodeB)**	Maximum high-precision backward jitter on the BSTWAMP
1542460412	**VS.BSTWAMP.Sender.TxPackets (LTE eNodeB, 5G gNodeB)**	Number of packets transmitted by the sender on the BSTWAMP
1542460413	**VS.BSTWAMP.Reflector.TxPackets (LTE eNodeB, 5G gNodeB)**	Number of packets transmitted by the reflector on the BSTWAMP
1542460414	**VS.BSTWAMP.Sender.RxPackets (LTE eNodeB, 5G gNodeB)**	Number of packets received by the sender on the BSTWAMP

Using the Real-Time Performance Monitoring Task

TWAMP supports real-time performance monitoring. When creating a real-time performance monitoring task, enable the TWAMP sender to send packets at an interval of 10 to 50 ms.

Perform the following operations to monitor TWAMP performance on the LMT in real time:

Step 1 On the LMT, click **Monitor**. The **Monitor** tab page is displayed.

Step 2 In the monitor navigation tree, choose **Monitor** > **Common Monitoring** > **Link Performance Monitoring**.

Step 3 In the displayed dialog box, enter the cabinet number, subrack number, and slot number, select **TWAMP SENDER**, specify the client index and sender index, and click **OK** to view the TWAMP measurement results.

----End

Perform the following operations to monitor TWAMP performance on the MAE in real time:

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 Choose **Base Station Device and Transport** > **Transport Performance Monitoring** > **Transport Link Traffic Monitoring**.

Step 3 In the displayed dialog box, set **Protocol Type** to **IP** and **Trace Type** to **TWAMP SENDER**, select the required monitoring items, specify the client index and sender index, and click **Finish**.

Step 4 Double-click the task to view the TWAMP measurement results.

----End

NOTE

If the TWAMP sender does not receive any packets in a measurement period due to severe packet loss in the forward or backward direction, the TWAMP sender cannot identify whether packets are lost in the forward or backward direction, and the measured forward and backward packet loss rates are inaccurate.

4.5.3 Network Monitoring

Checking Alarms

If the local end serves as the Controller, an NE reports alarms listed in **Table 4-21** when TWAMP activation fails.

- Full architecture

  After the TWAMP Controller function is enabled at the local end, the local end starts negotiations with the TWAMP Responder. If the negotiations are unsuccessful for four consecutive minutes, alarms listed in **Table 4-21** are reported.

  In a normal measurement process, the TCP keepalive timer on the control plane will be started every 2 hours, and keepalive packets are sent for up to nine times at an interval of 75s. That is, when no TCP packet is transmitted on the TCP connection for 2 hours, the TWAMP client sends a keepalive packet to the peer end. If the peer end does not respond, the base station continues to send the keepalive packet for up to eight times at an interval of 75s. If the peer end still does not respond after the keepalive packet is sent for nine times, the base station deletes the corresponding TCP connection. After the control-plane connection is interrupted due to no responses to the keepalive packets from the peer end for nine consecutive times, the local end initiates renegotiation. If the renegotiation is unsuccessful for four consecutive minutes, the alarms listed in **Table 4-21** are reported.

In a normal measurement process, if all the tests of the Control-Client are interrupted on the user plane and the local end receives no responses from the peer end for 15 consecutive minutes, the local end initiates renegotiation. If the renegotiation is unsuccessful for four consecutive minutes, alarms listed in **Table 4-21** are reported.

- Light architecture

  After the TWAMP Controller is configured, the TWAMP Controller sends test packets to the Responder. If the TWAMP Controller does not receive any response packets for four consecutive minutes, the alarms listed in **Table 4-21** are reported.

Table 4-21 Fault alarms related to IP Active Performance Measurement

RAT	NE	Alarm Name
GU	BSC/RNC (BSC6900/ BSC6910)	ALM-21354 IP Link Performance Measurement Fault
GU	BSC/RNC (BSC6960)	ALM-21662 IP Link Performance Measurement Fault
GULTN	eGBTS/NodeB/eNodeB/ gNodeB	ALM-25904 IP Link Performance Measurement Fault

Using MML Commands

Operators can use MML commands to troubleshoot faults and query the causes of activation failures.

- If an NE serves as the TWAMP Responder in full architecture, run the **DSP TWAMPRESPONDERSTA** command to check the negotiation statistics.
  - If the cause is transmission interruption, unavailable routing, incorrect configuration of the peer IP address, the Responder cannot receive any TCP negotiation packets. In this case, check **Number of TCP Setup Request Packets Received**.
  - If the local TCP resources, such as ports, are limited, any TCP connection establishment attempts will be rejected. In this case, check **Number of TCP Rejection Packets Sent**.
  - If the local end does not accept the mode of communication requested by the peer end, the local end replies with a NAK message in response to the received Set-Up-Response message. In this case, check **Number of Set-up-response Rejection Packets**.
  - If the local UDP resources, such as ports, are limited, the local end replies with a NAK message in response to the received Request-TW-Session message. In this case, check **Number of Request-TW-Session Rejection Packets Sent**.
  - If the peer end stops the test, the local end receives a Stop-Session message. In this case, check **Number of Stop-Session Packets Received**.

If the NE complies with RFC 5938, the local end receives a Stop-N-Sessions message instead of a Stop-Session message. In this case, check **Number of Stop-N-Sessions Packets Received**.

- If an NE serves as the TWAMP Responder in light architecture, run the **DSP TWAMPRESPONDER** command to query the statistics about TWAMP test sessions at the local end.
  - If the transmission is interrupted, routes are unavailable, or the peer IP address is incorrectly configured, the location information of the test session cannot be obtained. The location information refers to the source IP address, destination IP address, DSCP, source UDP port number, and destination UDP port number.
  - If the number of passive response test sessions is limited at the local end, the local end can only measure the location information of the number of test sessions within the specifications.
  - If the peer end stops measurement, the local end no longer receives any test packets from the peer end.
- If an NE serves as the TWAMP Controller in full architecture, run the **DSP TWAMPCLIENT** and **DSP TWAMPSENDER** commands to check the negotiation status and the negotiation failure cause, if any.

  For a base station:
  - If the cause is transmission interruption, unavailable routing, incorrect configuration of the peer IP address, the Controller cannot receive negotiation packets from the peer end. In this case, the value of **Negotiation Failure Cause** is **TCP_LINK_DOWN**.
  - If the peer TCP resources, such as ports, are limited, any TCP connection establishment attempts will be rejected. In this case, check whether **Negotiation Failure Cause** is **Server internal error**.
  - If the local end does not accept the mode of communication requested by the peer end, the local end replies with a NAK message in response to the received Set-Up-Response message. In this case, check whether **Negotiation Failure Cause** is **Server no support**.
  - If the peer UDP resources, such as ports, are limited, the peer end replies with a NAK message in response to the received Request-TW-Session message. In this case, the value of **Negotiation Failure Cause** is **Server resource limitation**.

  For a base station controller:
  - If the cause is transmission interruption, unavailable routing, incorrect configuration of the peer IP address, the local end cannot receive any responses to the negotiation packets. In this case, the value of **Status Change Cause** is **Connection expired**.
  - If the peer TCP resources, such as ports, are limited, any TCP connection establishment attempts will be rejected. In this case, the value of **Status Change Cause** is **Server internal error**.
  - If the local end does not accept the mode of communication requested by the peer end, the local end replies with a NAK message in response to the received Set-Up-Response message. In this case, the value of **Status Change Cause** is **Mode unsupported**.
  - If the peer UDP resources, such as ports, are limited, the peer end replies with a NAK message in response to the received Request-TW-Session

message. In this case, the value of **Status Change Cause** is **Temporary resource limitation**.

- If an NE serves as the TWAMP Controller in light architecture, run the **DSP TWAMPSENDER** command to query the statistics about TWAMP test sessions at the local end.

  If the transmission is interrupted, the route is unreachable, the peer IP address or UDP port number is incorrectly configured, or the specifications at the peer end are limited, the number of received packets of the session cannot be counted.
- Troubleshoot according to the related statistics on the TWAMP Responder end or the negotiation failure causes on the TWAMP Controller end.
  - Check the network connection if the peer end fails to respond in a specified time or does not respond at all.
  - Check whether TWAMP is enabled at the peer end, the configuration of the peer end is correct, and the specifications at the peer end are limited if the network is normal.
  - Check whether the negotiation failure cause is that resources on the peer end are limited. If yes, re-enable the TWAMP functions.
  - Check whether any transmission device prohibits the use of the ports (as described in **Table 4-15**) required for the TWAMP functions.

Fault Locating Method

Only the base station supports TWAMP detailed tracing on the MAE. Currently, this tracing function can be used to locate control packet issues and test packet issues in TWAMP full architecture in IPv4 and IPv6 transmission and locate test packet issues in TWAMP light architecture in IPv4 and IPv6 transmission.

1. On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.
2. Choose **Base Station Device and Transport** > **Information Collection** > **TWAMP Protocol Detail Trace**.
3. In the displayed dialog box, set **TWAMP protocol type** to **TWAMP-Control(TCP)** for the location of control packet issues, set **TWAMP protocol type** to **TWAMP-Test(UDP)** for the location of test packet issues, or set **TWAMP protocol type** to **ALL** for the location of both control packet and test packet issues.
4. In the displayed dialog box, set **TWAMP Object Type** to **Client** and specify the client index if the local end serves as the TWAMP controller, set **TWAMP Object Type** to **Responder** and specify the responder index if the local end serves as the TWAMP Responder, or set **TWAMP Object Type** to **ALL** if the local end serves both as the TWAMP Controller and Responder.
5. Set **Trace direction**, and click **Submit** to view the tracing results.

- Locating control packet issues
  - Check the tracing results. The control messages are sent in the following order: server-greeting, set-up-response, server-start, Request-TW-Session, Accept-Session, Start-Sessions, Start-Ack, and Stop-Sessions. If the NE complies with RFC 5938, the control messages are sent in the following order: server-greeting, set-up-response, server-start, Request-TW-Session,

Accept-Session, Start-N-Sessions, Start-N-Ack, Stop-N-Session, and Stop-N-Ack.

- Check whether the local end has transmitted the expected packets or has received the expected packets from the peer end.
  - Check the network connection if the peer end fails to respond in a specified time or does not respond at all.
  - If the local end has not transmitted any expected packets, run the **DSP TWAMPCLIENT** and **DSP TWAMPSENDER** commands to check the failure cause.
  - If the failure cause is that resources at the peer end are limited, troubleshoot the peer end and re-enable the TWAMP functions.
  - Check whether any transmission device prohibits the use of the ports (as described in **Table 4-15**) required for the TWAMP functions.
- Check whether the local end and peer end exchange TCP packets as expected. For details about the TCP packets exchanged between the two ends, see RFC 5938 (if individual session control is enabled) or RFC 5357.

- Locating test packet issues
  - View the TWAMP tracing results.
  - Check whether the local end has transmitted the expected packets or has received the expected packets from the peer end.
    - Check the network connection if the peer end fails to respond in a specified time or does not respond at all.
    - Check whether any transmission device prohibits the use of the ports (as described in **Table 4-15**) required for the TWAMP functions.
  - Check whether the local end and peer end exchange UDP packets as expected. For details about the UDP packets exchanged between the two ends, see RFC 5357 or RFC 5938.

5 RFC 2544 Tests

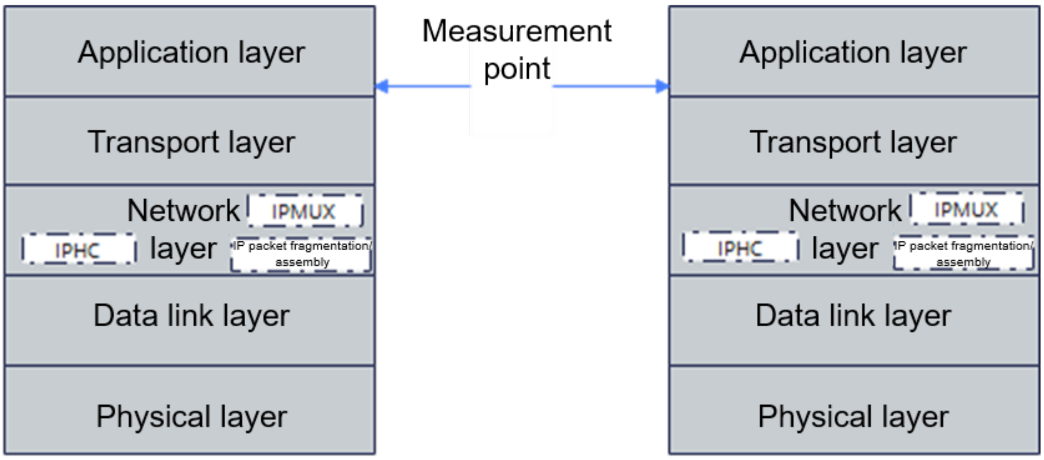
5.1 Principles

5.1.1 Basic Principles

As defined by the IETF, RFC 2544 provides general traffic testing standards and the benchmarking methodology for evaluating network performance, which can be used to describe the performance characteristics of a network interconnecting device. In accordance with RFC 2544, E2E testing is implemented through packet self-sending with the use of testers or RFC 2544–compliant devices, regardless of whether services have been deployed on networks. Operators can measure the performance of the current transmission link through RFC 2544 to check whether the transport network can meet service requirements for service admission optimization.

In RFC 2544 tests, measurements are performed above the transport layer in the TCP/IP protocol model, as shown in **Figure 5-1**.

Figure 5-1 Measurement point for RFC 2544 tests in the TCP/IP protocol model



A device involved in an RFC 2544 test may be an initiator or reflector. A wireless base station can only serve as a reflector.

- Initiator: proactively sends packets to simulate service traffic for testing.
- Reflector: receives test packets from the initiator and loops back the packets to the initiator.

In an IPv4- or IPv6-based RFC 2544 test, the base station can only serve as the reflector to loop back packets from the initiator. The test result needs to be obtained from the initiator.

RFC 2544 defines multiple transmission quality measurement indicators, which can be used to evaluate the performance of a customer's transport network. The indicators are as follows:

- Throughput: indicates the fastest rate at which packets are transmitted without packet loss on a link.
- Two-way packet loss rate: indicates the percentage of discarded packets on a link during E2E forwarding.
- Two-way delay: indicates the duration from the time when packets are sent from the initiator to the time when packets are looped back to the initiator.

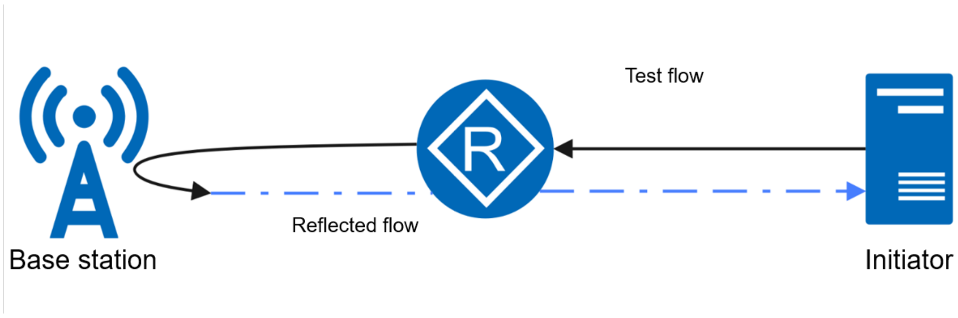
5.1.2 Application Scenario

Currently, a base station supports only the RFC 2544-compliant network layer testing and does not support the data link layer testing. The IP addresses and UDP port numbers of the two ends involved in a test must be specified. Packet tests in non-security scenarios and security scenarios are both supported.

Packet Test in Non-Security Scenarios

In non-security scenarios, the base station serves as the reflector to loop back packets sent by a remote device that can serve as the initiator defined by RFC 2544. After test packets are captured by the base station, the base station matches the IP address and UDP port number configured on the reflector, exchanges the source IP address and UDP port number of the packets with the destination IP address and UDP port number of the packets, and searches for a route to send the packets. **Figure 5-2** shows the test flow loopback in non-security scenarios.

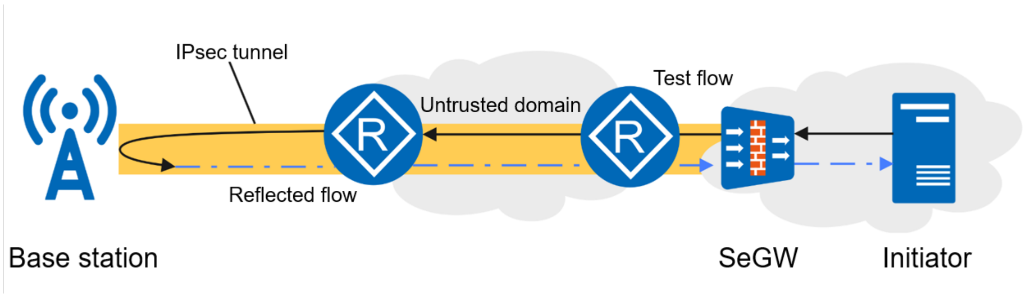
Figure 5-2 Test flow loopback in non-security scenarios



Packet Test in Security Scenarios

In security scenarios, an IPsec tunnel is established between the base station and the SeGW. The test packets are sent from the initiator to the base station through the IPsec tunnel. After verifying and decrypting the test packets, the base station reflects the packets based on the reflector configuration. In this way, IPsec-protected RFC 2544–compliant measurement is implemented. **Figure 5-3** shows the test flow loopback in security scenarios.

Figure 5-3 Test flow loopback in security scenarios



5.2 Network Analysis

5.2.1 Benefits

RFC 2544 tests help operators learn about the actual transmission capability of the network and decide on service admission accordingly, ensuring stable network operation and reducing network operation costs.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

- In RFC 2544 tests, UDP packets are used. The base station serves as the reflector in the standard tests. Packet processing consumes the user-plane forwarding performance resources of the base station. A throughput test consumes a large number of user-plane forwarding performance resources. Therefore, you are not advised to perform an RFC 2544 test when service traffic is heavy. Otherwise, services may be affected.
- The RFC 2544 throughput test involves active packet injection, so traffic needs to be injected into the network. The consumed bandwidth depends on the rate at which the initiator sends test packets.

5.3 Requirements

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.1 Licenses

There are no license requirements for basic functions.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

Only the UMPTha, UMPThb, and UMPTg boards are supported, and the maximum board bandwidth compatible with RFC 2544 throughput tests is 10 Gbit/s.

RF Modules

No requirements

5.3.4 Others

- If a base station serves as the reflector in an RFC 2544 test, **UDPLOOP**.***LM*** must be set to **SPECIALIPUDPPORTREMOTE**. Modes indicated by other values are used only for connectivity tests.
- If **UDPLOOP**.**LM** is set to **SPECIALIPUDPPORTREMOTE**, the destination IP address of the local end must have been configured on the base station so that packets can be correctly looped back.
- The RFC 2544 throughput test involves packet injection, which affects existing services on the link. In return, the existing service traffic affects the loopback performance of the base station. It is recommended that a throughput test be

performed when the network traffic is light or when services are not accessed, for example, when a new transport network is deployed or a transport network is swapped or reconstructed. This ensures measurement accuracy and avoids service loss. In addition, it is recommended that this feature be deactivated after the test is complete.

- Regarding throughput tests, it is recommended that the maximum test rate should not exceed the maximum reflection capability of a board. Otherwise, the link bottleneck may fail to be accurately tested.
- Packet tracing affects the reflective performance of the base station. Therefore, you are advised not to enable packet tracing during throughput tests.
- If Chinese cryptographic algorithms or weak algorithms are used for security scenarios, encryption and decryption have a great impact on performance, and the reflective performance deteriorates.
- This feature does not apply to packet fragmentation scenarios. As a result, the length of test packets sent by the initiator must be less than or equal to the minimum maximum transmission unit (MTU) of the link to prevent packets from being fragmented during tests.
- Only one **UDPLOOP** MO can be configured for a single base station. If another **UDPLOOP** MO is to be configured, the current MO will be overwritten.
- The **UDPLOOP** MO where ***APPTIF*** is configured cannot be configured with the **SRCIPRT** MO at the same time.
- When **UDPLOOP**.***LM*** is set to **SPECIALIPUDPPORTREMOTE**, the precautions are as follows:
  - **UDPLOOP**.***RXSPN*** cannot be set to a value ranging from **62048** to **62559**.
  - When **UDPLOOP**.***RXDPN*** is set to **65040** and the IP address specified by **UDPLOOP**.***RXDIP*** is the same as that specified by **USERPLANEHOST**.***LOCIPV4*** in endpoint mode, link environment loopback detection in endpoint mode may be abnormal.
  - If both the IP port and UDP port match the actual service packets in the same routing domain, services will be interrupted. Therefore, exercise caution during configuration.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-1 describes the data to be prepared for configuring a base station as the reflector in an RFC 2544 test.

Table 5-1 Data to be prepared for configuring a base station as the reflector in an RFC 2544 test

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	**UDPLOOP**.***CN***	Set this parameter based on the network plan.
Subrack No.	**UDPLOOP**.***SRN***	Set this parameter based on the network plan.
Slot No.	**UDPLOOP**.***SN***	Set this parameter based on the network plan.
Loopback Mode	**UDPLOOP**.***LM***	If the base station serves as the reflector in an RFC 2544 test, set this parameter to **SPECIALIPUDPPORTREMOTE**.
VRF Index	**UDPLOOP**.***VRFINDEX***	Set this parameter based on the network plan.
IP Version	**UDPLOOP**.***IPVERSION***	Set this parameter based on the network plan.
RX Source IP Address	**UDPLOOP**.***RXSIP***	Set this parameter based on the network plan.
RX Destination IP Address	**UDPLOOP**.***RXDIP***	Set this parameter based on the network plan.
TX Source IP Address	**UDPLOOP**.***TXSIP***	Set this parameter based on the network plan.
TX Destination IP Address	**UDPLOOP**.***TXDIP***	Set this parameter based on the network plan.
RX Source IPv6 Address	**UDPLOOP**.***RXSIP6***	Set this parameter based on the network plan.
RX Destination IPv6 Address	**UDPLOOP**.***RXDIP6***	Set this parameter based on the network plan.
TX Source IPv6 Address	**UDPLOOP**.***TXSIP6***	Set this parameter based on the network plan.
TX Destination IPv6 Address	**UDPLOOP**.***TXDIP6***	Set this parameter based on the network plan.
RX Source Port Number	**UDPLOOP**.***RXSPN***	Set this parameter based on the network plan.
RX Destination Port Number	**UDPLOOP**.***RXDPN***	Set this parameter based on the network plan.
TX Source Port Number	**UDPLOOP**.***TXSPN***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
TX Destination Port Number	**UDPLOOP.*TXDPN***	Set this parameter based on the network plan.
Loopback Time	**UDPLOOP.*LT***	You are advised to set this parameter to a value greater than the duration when the initiator is involved in an RFC 2544 test.
Appoint Interface	**UDPLOOP.*APPTIF***	Set this parameter when **IP Version** is set to **IPv4**. The default value is **NO**.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring the UDPLOOP MO for the base station, and setting the IP version to IPv4
SET UDPLOOP: CN=0, SRN=0, SN=7, LM=SPECIALIPUDPPORTREMOTE, IPVERSION=IPv4,
RXSIP="172.16.81.122", RXDIP="172.16.81.123", RXSPN=1111, RXDPN=1025, TXSIP="172.16.81.124",
TXDIP="172.16.81.125", TXSPN=1025, TXDPN=1111, LT=1, APPTIF=NO;
//Configuring the UDPLOOP MO for the base station, and setting the IP version to IPv6
SET UDPLOOP: CN=0, SRN=0, SN=7, LM=SPECIALIPUDPPORTREMOTE, IPVERSION=IPv6,
RXSIP6="2015:1960::123", RXDIP6="2015:240::11", RXSPN=1111, RXDPN=1025, TXSIP6="2015:240::11",
TXDIP6="2015:1960::123", TXSPN=1025, TXDPN=1111, LT=1;
```

Deactivation Command Examples

```
//Configuring the UDPLOOP.LM parameter for the base station and deactivating the RFC 2544 test
SET UDPLOOP: CN=0, SRN=0, SN=7, LM=NOLOOP;
```

5.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

The base station serves only as the reflector in an RFC 2544 test, and the test result needs to be observed on the initiator.

Using Signaling Tracing

Start IP layer protocol tracing to trace the destination IP address of the current reflection source. Use the initiator to send test packets to the base station. If reflected packets are observed in the base station tracing result, the activation succeeds.

Command Query

Run the **DSP UDPLOOP** command to check the currently configured reflective state. If the value of **Loopback Mode** is **Special IP and UDP Port Remote Loopback**, the activation succeeds.

5.4.3 Network Monitoring

None

6 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- *SRAN Networking and Evolution Overview*
- *IPv4 Transmission*
- *IP Performance Monitor*
- *Transmission Resource Pool in BSC* in *GBSS Feature Documentation*
- *Transmission Resource Pool in RNC* in *RAN Feature Documentation*